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ABBREVIATIONS USED IN THE PAPER

ARIMA - Autoregressive Integrated Moving Average

ARIMAX - Autoregressive Integrated Moving Average with Exogenous Variables

HVAC - Heating, Ventilation, and Air Conditioning

KPI - Key Performance Indicator

LSTM - Long Short-Term Memory

MAE - Mean Absolute Error

MPC - Model Predictive Control

RMSE - Root Mean Squared Error

SARIMA - Seasonal Autoregressive Integrated Moving Average

SARIMAX - Seasonal Autoregressive Integrated Moving Average with Exogenous Variables

INTRODUCTION

Cooling systems are a critical component of telecom central offices, where stable environmental conditions are required for the reliable and continuous operation of sensitive electronic equipment. In such environments, cooling is not a comfort-driven service but an operational necessity, and its energy use forms a substantial share of total site demand. Telecom-specific studies and adjacent data-centre research show that supporting systems, especially cooling, are a major component of infrastructure electricity consumption (Sorrentino *et al.*, 2019; D'Aniello *et al.*, 2018). As a result, cooling performance is not a secondary technical issue, but an important part of site-level energy management.

Modern telecom facilities generate large volumes of monitoring data, including subsystem energy signals, environmental variables, and operational indicators. However, these data are mostly used for monitoring, visualisation, and diagnostics rather than for predictive or supervisory decision-making. At the same time, improving energy efficiency has become increasingly important under cost, sustainability, and reliability constraints. This creates a practical need for analytical approaches that can transform existing telemetry into defensible operational insight without compromising thermal safety.

A further challenge lies in the nature of the available data. In real monitored infrastructure, cooling demand is often not available as a ready-made hourly variable. Instead, the analytical target may need to be reconstructed from fragmented cumulative subsystem measurements, incomplete contextual signals, and uneven data quality. The reviewed literature shows that telecom studies are comparatively well developed in monitoring, KPI design, and diagnosis, but less developed in the full chain from target engineering to short-horizon cooling-demand forecasting, expected-versus-actual benchmarking, and forecast-informed supervisory control under sensing constraints (Qaisar and Zhao, 2022; Malafronte, Sorrentino and Trifirò, 2021; Mea *et al.*, 2017; Sorrentino *et al.*, 2016). The present thesis addresses this gap by combining these strands within one real telecom central office case.

The aim of the thesis is to develop and evaluate intelligent control strategies for cooling energy use in telecom central office environments using real monitoring data. The aim is pursued through the construction of a control-oriented analytical basis from fragmented telemetry, the development of a short-term expected-demand benchmark, and the historical comparative evaluation of intelligent control strategies.

Accordingly, the thesis is positioned as an applied study of intelligent cooling control rather than as full model predictive control deployment, reinforcement-learning control, or actuator-level closed-loop validation. In this thesis, intelligent control strategies are evaluated on historical operation through benchmark-based and context-aware decision logic rather than as deployed closed-loop controllers. This scope is methodologically consistent with the available telemetry and with the research gap identified in the literature review.

To achieve this aim, the thesis solves the following research tasks:

1. To conduct a literature review on cooling analytics, predictive benchmarking, and intelligent control strategies.
2. To explore, preprocess, and structure real operational and environmental data into a control-oriented analytical basis.
3. To develop a reliable short-term forecasting benchmark for expected cooling demand.
4. To design intelligent cooling control strategies using forecast-based expected-demand signals and operational context.
5. To evaluate the proposed control strategies through historical comparative evaluation.

The research questions of the thesis are:

1. To what extent can intelligent cooling control strategies identify opportunities to improve cooling energy use in telecom central office environments?
2. Which forecasting approach provides the most reliable estimate of expected cooling demand for intelligent cooling control?
3. Which operational and environmental variables are most relevant for constructing a control-oriented analytical basis for intelligent cooling control?

The object of the research is cooling energy use in telecom central office technical infrastructure.

The subject of the research is intelligent cooling control in telecom central office environments under real operational telemetry constraints.

The methods used in the thesis are:

1. Documented literature search and structured literature review on cooling analytics, predictive benchmarking, and intelligent control strategies in technical-infrastructure environments.

2. Preprocessing, exploratory analysis, signal screening, and target engineering of real operational and environmental data using Python-based analytical tools to construct a control-oriented analytical basis.
3. Development, comparison, and validation of short-term forecasting models using time-series and machine-learning techniques in order to estimate expected cooling demand.
4. Forecast-based operational benchmarking through comparison of actual and expected cooling demand as the decision layer for intelligent control.
5. Historical comparative evaluation and sensitivity-based assessment of intelligent cooling control strategies.

The expected result of the thesis is a defensible analytical framework for telecom cooling analytics that integrates engineered cooling-demand telemetry, short-horizon expected-demand modelling, benchmark-based operational assessment, and intelligent control evaluation. The framework is intended to support the identification of context-aware opportunities for improving cooling energy use.

The structure of the thesis can be described in a following way. Chapter 1 presents the literature review and establishes the theoretical and methodological background of the study, covering telecom cooling in technical infrastructure, operational data quality and target engineering, short-term cooling-demand forecasting, intelligent control strategies, and benchmark-based evaluation. Chapter 2 describes the research design, case-study setting, data sources, signal screening, engineering of the hourly cooling-demand target, dataset construction, exploratory data analysis, forecasting methodology, and the intelligent control methodology formulated as a simplified sequential decision framework. Chapter 3 presents the results and analysis of the study. It begins with expected-demand benchmark selection, then proceeds to benchmark-based cooling assessment, the transition from benchmark deviation to candidate control hours, comparative results for the strategy families, operating modes and risk states, and the sensitivity and robustness analysis of the control framework. Chapter 4 discusses the main findings, answers the research questions, states the contribution of the thesis, and identifies limitations and directions for future research. The thesis ends with conclusions, the list of references and sources, and annexes.

1. LITERATURE REVIEW

1.1 Literature search strategy, scope and bibliometrics

This literature review was guided by a documented search strategy designed to identify studies relevant to intelligent control of cooling energy use in telecom central office environments, together with the data-engineering and forecasting methods required to support such control under real operational constraints. The review focused on five thematic areas that structure the thesis which are telecom cooling diagnostics, operational data preprocessing and target engineering, short-term cooling-demand forecasting, intelligent control strategies for cooling systems, and benchmark-based evaluation. Because telecom-specific evidence remains limited, relevant transfer literature from data centres and building HVAC research was also included when it contributed directly to the methodological framing of intelligent control, expected-demand estimation, or benchmark-based assessment under operational and thermal constraints.

Search terms were organised into concept groups covering domain, system, outcome, method, and operational context. The domain group included telecom*, "telecommunication central office", "data center", datacenter, "server room", and "mission-critical". The system group included cool*, HVAC, "air condition*", and chiller*. The outcome group included "energy consumption", "cooling load", "power demand", "energy efficien*", "baseline", and "anomaly detection". The method group included forecast*, prediction, "time series", ARIMA, LSTM, "gradient boosting", "random forest", "model predictive control", MPC, "rule-based", and "reinforcement learning". These groups were combined using Boolean OR within groups and AND between groups. The main databases were Scopus and Google Scholar. Scopus was used for structured filtering and metadata control, while Google Scholar was used to broaden recall and support backward and forward citation chaining. AI-assisted tools were used only to suggest query variants. All inclusion decisions were verified against the original source (Martín-Martín *et al.*, 2018).

Studies were included when they were available in full text, written in English, and published mainly between 2015 and 2025, while older seminal studies were retained only when methodologically necessary. Priority was given to peer-reviewed empirical studies, comparative model-evaluation studies, and review papers that provided essential theoretical framing. Eligible studies addressed cooling energy use, cooling-load forecasting, baseline or anomaly modelling, or intelligent control of cooling and HVAC systems in telecom, data-centre, or closely related technical-infrastructure settings. Studies were excluded if they focused only on heating, lacked direct relevance to cooling-energy analysis, were purely descriptive without analytical or methodological contribution, duplicated a more complete version of the same study, or had inaccessible full text. The review question, keyword design, database selection, and screening logic were defined in advance to improve transparency and reproducibility (Page *et al.*, 2021; Okoli, 2015).

The search process combined database searching with backward and forward citation tracking from core papers. Records were screened using title and abstract for relevance and were then assessed as a full text against the inclusion and exclusion criteria. The final set of studies was organised thematically according to the structure of this thesis. This strategy was designed to balance breadth, methodological relevance, and reproducibility for a real-data telecom cooling study, while positioning forecasting and target engineering as supporting methodological layers for the broader problem of intelligent cooling control.

A complementary bibliometric overview was used to describe the broader structure of the reviewed literature. The organised Zotero database contained over 35 peer-reviewed publications and showed a visible increase in publication activity after 2021. Keyword frequency and co-occurrence patterns indicated strong thematic concentration around machine learning, HVAC, energy efficiency, predictive control, and intelligent control, while journal analysis showed that the literature is concentrated in established energy and building-science venues. This overview supports the thematic organisation of the review and confirms that forecasting, benchmarking, and intelligent control are closely connected within the current research landscape.

1.2. Cooling energy in telecom central offices and related technical infrastructure

Telecommunication central offices are highly energy-intensive technical environments in which cooling is an operational necessity rather than a comfort-driven service. In such sites, total electricity use is typically divided into direct consumption associated with telecommunication equipment and indirect consumption associated mainly with supporting systems, especially cooling. Telecom-specific studies indicate that this indirect share can become very large, in some cases approaching half of total grid-supplied energy, which makes cooling performance a critical part of site-level energy management rather than a secondary auxiliary issue (Sorrentino *et al.*, 2019; D'Aniello *et al.*, 2018). This broader importance is consistent with adjacent data-centre literature, where air-conditioning systems account on average for about 38% of facility energy use and are therefore a major target for efficiency assessment and benchmarking (Sorrentino *et al.*, 2019).

The telecom literature has approached this problem primarily through monitoring, diagnostics and energy-intelligence frameworks. Sorrentino et al. (2016) developed a approach which is model based for monitoring and diagnosis of telecommunication cooling systems, while Mea et al. (2017) extended the same line of work toward real-time monitoring, diagnosis and offline optimisation analysis at room level. D'Aniello et al. (2018) proposed high-level energy performance metrics for telecommunication sites, including indicators explicitly designed to identify anomalous energy absorption and data reliability problems. Malafronte et al. (2021) further showed that meaningful diagnosis can be achieved largely from existing electricity measurements, without major expansion of the sensing infrastructure and reported measurable yearly savings in a high-demand central office after deployment of the proposed diagnostic logic.

A consistent theme in this telecom literature is the use of synthetic, benchmark-oriented indicators to translate complex operational data into interpretable diagnostic measures. D'Aniello et al. (2018) combine reliability- and benchmark-oriented indices to distinguish unreliable measurements from genuinely anomalous site behaviour, while Sorrentino et al. (2019) develop the utilization factor as a climate-independent metric tailored to the energy breakdown of telecommunication central offices. Malafronte et al. (2021) extend this direction by showing that diagnosis-oriented energy intelligence can be built largely from existing electricity measurements, without major expansion of the sensing infrastructure. Together, these studies frame telecom cooling not simply as a load to be monitored, but as an operational process to be assessed against reference behaviour using data-driven diagnostic logic.

This telecom KPI and diagnosis line has also been extended into longer-horizon energy-management support. Sorrentino et al. (2024) develop a multi-annual monitoring tool for telecommunication central offices that classifies sites by efficiency and measurement reliability using KPI-based analysis over multi-year periods. This is useful for the present thesis because it strengthens the argument that telecom cooling analytics are not limited to one-off diagnosis, but belong to a broader energy-intelligence tradition in which benchmark-oriented monitoring, reliability assessment and operational interpretation are tightly connected.

Within the reviewed telecom corpus, monitoring, KPI development and diagnosis appear more developed than short-horizon cooling-demand forecasting from fragmented operational telemetry. Existing studies show clearly why cooling energy should be monitored, benchmarked and diagnostically interpreted in telecommunication sites, but they provide less direct support for the full chain from target reconstruction to short-term forecasting and model-based expected-demand benchmarking. This is the point at which the present thesis is positioned. It builds on the telecom tradition of energy intelligence and benchmark-oriented diagnosis, but extends it toward data-driven forecasting and operational assessment of cooling demand in a real central-office environment.

1.3 Operational data quality, preprocessing and target engineering

A repeating finding in the energy of the building literature is that operational data are rarely ready for direct modelling. Fan et al. (2021) describe data preprocessing as an indispensable stage of discovery of the known facts from operational data of the certain building and show that missing values, outliers, reduction, scaling, transformation and partitioning can all materially affect subsequent analysis. They also stress that preprocessing in real buildings cannot be fully standardised because both data quality and operating characteristics vary across applications. In real industrial telemetry, preprocessing choices also determine whether later forecasting and benchmarking results reflect genuine operational structure or artefacts of missingness, misalignment and sensor instability.

In practice, however, monitored datasets rarely provide a complete and uniformly reliable representation of all relevant variables. The preprocessing literature therefore supports a pragmatic distinction between signals that can be used directly, signals that require transformation and signals that should be excluded because of instability or insufficient completeness. This is particularly important in operational energy analytics, where the validity of later benchmarking depends on whether the retained variables reflect persistent system behaviour rather than defects in data collection or storage.

The most relevant methodological literature for the present thesis is the literature on sub-metering. Li et al. (2023) argue that sub-metering is a main point for evaluation of level of energy efficiency because total building energy consumption is too aggregated to reveal the behaviour and savings potential of individual sub-systems. Their study is especially important because it treats sub-item prediction and sub-item evaluation as analytically more informative than total-consumption assessment alone. In this view, itemized energy data are not simply a richer dataset, but a different level of observation that allows operational patterns to be assigned to specific end uses rather than being diluted in a whole-building total.

The difficulty of obtaining a usable cooling target is also reflected in submetering-based HVAC assessment studies. Ji et al. (2017) combine electricity submetering with cooling-load estimation in order to assess main HVAC equipment performance, while Li et al. (2023) distinguish direct and indirect metering relationships when reconstructing sub-item energy behaviour. Together, these studies support the view that cooling demand may need to be inferred from distributed measurements rather than directly observed as a single ready-made variable. This is particularly relevant in monitored technical infrastructure, where subsystem-level cumulative meters may be the only consistent trace of cooling activity. In such cases, the metering layer and the analytical target are not equivalent: cumulative subsystem measurements must first be converted into interval consumption through meter selection, differencing, treatment of physically implausible negative steps, aggregation and outlier control before they can function as an hourly cooling-demand series.

For benchmark-oriented modelling, however, predictive usefulness is not the only criterion for input selection. Qaisar and Zhao (2022) stress that the meaning of a baseline depends on the chosen independent variables and Chiosa et al. (2022) similarly show that anomaly detection depends on both a robust reference model and an appropriate definition of the features used to identify deviation from normal behaviour. In this setting, variables drawn from the same signal family as the target may improve fit while also reducing interpretability and increasing circularity risk. This issue is especially relevant when target-adjacent cooling proxies are available since they may strengthen predictive accuracy numerically, yet weaken the independence between explanatory inputs and the reconstructed target that the model is intended to benchmark.

A closely related strand is limited-metering benchmarking and abnormal-consumption detection. Canale et al. (2024) define abnormal thermal energy consumption as deviation from a reference value and show that diagnosis remains possible even when metering infrastructure is incomplete, provided that benchmarking is performed systematically and contextual variables are incorporated appropriately. Their study is particularly useful for the present thesis because it frames limited instrumentation not as a reason to abandon analysis, but as a reason to design the analytical pipeline more carefully. It also reinforces the idea that reference-based interpretation is essential. Abnormality cannot be read directly from raw consumption values, but emerges only through comparison with an appropriate benchmark.

This logic also connects naturally to baseline modelling and measurement and verification. Qaisar and Zhao (2022) describe the energy baseline model as a reference estimate used to evaluate performance under changing conditions, while Alrobaie and Krarti (2022) show that data-driven baseline models are increasingly used for M&V applications when historical operational data are available. In the present thesis, this bridge is important because target engineering is not an end in itself. It matters because the reconstructed cooling series is later used not only for forecasting, but also as the basis for expected-versus-actual operational benchmarking.

Taken together, the literature supports treating target engineering as a substantive part of the thesis contribution rather than as a hidden preprocessing detail. In many forecasting studies, the target is assumed to be directly available as hourly load, temperature, or total energy demand. In the present case, by contrast, the hourly cooling target must be identified, screened, reconstructed from fragmented cumulative subsystem measurements and cleaned before forecasting can even begin. The literature on preprocessing, sub-metering, limited metering and baseline-oriented modelling shows that this is not a peripheral inconvenience but a central methodological issue in real-world energy analytics.

1.4. Short-term cooling demand forecasting as expected-demand estimation

Short-term cooling-demand forecasting is typically approached through three broad modelling families: physics-based models, data-driven models, and hybrid or grey-box models. For real operational datasets, data-driven approaches are often preferred because they can be trained directly on measured telemetry without requiring a fully specified physical description of the building or cooling system. Chen et al. (2022) and Khalil et al. (2022) both show that model suitability depends on the forecasting objective, the available system information, the temporal resolution of the data, and the intended use of the model. In the present thesis, this question is not treated as forecasting alone, but as part of the broader problem of constructing a reliable expected-demand reference for intelligent cooling control. In the present thesis, this problem is not treated as forecasting alone, but as part of the broader task of constructing a reliable estimate of expected cooling demand for intelligent control. In this context, a comparative modelling strategy is methodologically more appropriate than assuming in advance that either statistical models or deep-learning models should dominate.

The evaluation design is also a methodological choice rather than a neutral reporting detail. Mariano-Hernández et al. (2021) introduce a backtesting stage before final model selection, while Alrobaie and Krarti (2022) show that conclusions from data-driven energy models depend not only on the modelling algorithm but also on the adopted validation framework. For operational time series, this implies that chronological holdout testing alone may provide weaker evidence than rolling or expanding-window evaluation, which better reflects changing temporal conditions and repeated forecast deployment. The same logic also supports reporting more than one error metric. In applied energy forecasting, MAE and RMSE do not emphasise the same aspects of predictive performance, since RMSE penalises larger deviations more strongly, while MAE gives a more direct measure of average forecast error. As a result, comparative conclusions can differ depending on the chosen metric and both measures are useful when models are later interpreted operationally rather than only statistically.

Within this broader field, the most relevant literature for the present thesis is the cooling-load forecasting literature. Guo et al. (2014) model hourly cooling load using time-indexed autoregressive models with exogenous inputs and explicitly treat cooling demand as a function of both recent load history and external weather conditions. Their formulation is particularly important because it reflects the inertia of large cooling systems. Demand does not respond instantaneously to changing external conditions and recent load values therefore contain essential predictive information. The same study also shows that weather variables such as outdoor temperature and relative humidity remain relevant, but that their influence is filtered through system memory and varies with forecast horizon.

A second important strand concerns nonlinear machine-learning and deep-learning methods. Fan et al. (2017) show that nonlinear approaches are effective for short-term building cooling-load prediction and that deep learning can improve performance, especially when used for feature extraction. At the same time, their results do not support the assumption that supervised deep models automatically dominate simpler alternatives. Earlier next-day work by Fan et al. (2014) also shows that ensemble approaches can outperform individual predictors, while tree-based and other nonlinear models remain among the strongest individual methods in building-energy forecasting. A similar pattern appears in technical-infrastructure transfer literature. Kula et al. (2025), working with a monitored data-centre cooling system, compare modern neural architectures with Random Forest, XGBoost and AutoARIMA and show that tree-based methods remain highly competitive even when newer neural approaches are included. Taken together, these studies suggest that in cooling-related operational datasets, nonlinear tree-based models deserve to be treated as first-class benchmarks rather than as secondary baselines behind recurrent neural networks.

Predictor design is equally important. Prior studies identify historical demand, weather variables, and contextual features as core inputs in data-driven cooling forecasting (Khalil *et al.*, 2022; Mariano-Hernández *et al.*, 2021). At the same time, practical variable selection is often constrained by missingness and incomplete telemetry rather than by theoretical completeness (Chalapathy, Khoa and Sethuvenkatraman, 2021; Lin *et al.*, 2022). This is directly relevant to the present thesis, where dense indoor sensing is unavailable, but limited internal temperature information can still be interpreted as a slow thermal-context proxy alongside external weather and operational variables. As a result, the benchmark model may represent cooling demand sufficiently well for control-oriented analysis without directly observing the full indoor thermal state. However, when predictors are closely linked to the target-construction chain, higher predictive accuracy may come at the cost of weaker independence between drivers and response.

Forecast horizon changes the nature of the problem. Prior studies show that very short-horizon cooling prediction depends mainly on recent load history, whereas longer horizons rely more on broader temporal structure and exogenous inputs (Chalapathy, Khoa and Sethuvenkatraman, 2021; Guo *et al.*, 2014). This also explains why naive persistence remains a serious benchmark rather than a trivial one. Strong one-hour-ahead performance reflects the structure of the problem itself, not weak modelling design. In the present thesis, this supports the use of short-horizon forecasting as an expected-demand benchmark for later control-oriented interpretation rather than as an end in itself.

Overall, the literature supports a horizon-sensitive interpretation of cooling-demand forecasting. At very short horizons, cooling demand is typically dominated by recent history, whereas at longer horizons the relative importance of exogenous structure and model form may increase. In the present thesis, the methodological focus is on short-horizon hourly forecasting because the resulting model is used to estimate expected cooling demand, which then becomes the basis for benchmark-based interpretation and intelligent control logic. Forecasting is therefore not treated here as an end in itself, but as a methodological layer that supports intelligent cooling control under real telemetry constraints.

1.5. Predictive and intelligent control strategies for cooling systems

As shown in Table 1, the control methodology adopted in this thesis is best positioned as a simplified predictive and intelligent control approach. Control decisions are informed by forecast outputs and contextual operating information rather than by fixed schedules alone or by full online optimization. In the present study, the forecast model is used to identify candidate intervention periods, and bounded control actions are triggered only when additional operational conditions are satisfied. The thesis therefore does not implement full MPC, data-driven MPC, or model-free control, but instead applies a simplified control strategy consistent with the available telecom telemetry and the scope of historical comparative evaluation.

The reviewed control approaches ranged from rule-based control to predictive, data-driven, and model-free control strategies. The detailed comparison is provided in Annex 1. Given the available telemetry, the thesis adopts a simplified forecast-informed supervisory control approach rather than full MPC or deployed closed-loop control.

In the literature, intelligent control of HVAC and cooling systems is most commonly framed as a predictive-control problem. Reviews by Killian and Kozek (2016), Drgoňa et al. (2020), and Xin et al. (2024) show that predictive and constraint-aware decision-making remains the dominant mature paradigm, while more data-driven and model-free approaches continue to develop. A related conceptual perspective comes from sequential decision theory, where control problems are described through states, actions, transitions, and rewards (Sutton and Barto, 2018). This perspective is relevant here because the proposed control logic also operates through explicit control states, bounded actions, and decision rules over time, although the thesis does not formulate a full Markov decision process or learn an optimal policy.

A second strand of the literature focuses on reducing the practical barriers of advanced control through data-driven and approximate predictive-control approaches. Studies such as Maddalena et al. (2020), Kathirgamanathan et al. (2021), Yang et al. (2021), and Klaučo et al. (2014) show that simplified or learned predictive-control logic can preserve much of the value of advanced control while reducing modelling and implementation burden. This is relevant to the present thesis because it supports a simplified interpretation of intelligent control rather than restricting the term to full MPC deployment.

In data-limited supervisory settings, operating-mode discovery from historical telemetry may provide a practical alternative to direct plant-state modelling. Internal clustering criteria such as silhouette width and the Davies-Bouldin index are useful for assessing candidate partitions, but their interpretation remains strongest when combined with operational meaning rather than treated as decisive on their own (Arbelaitz *et al.*, 2013; Rousseeuw, 1987; Davies and Bouldin, 1979).

This bridge is particularly relevant when the forecasting layer is strongest in tree-based machine learning rather than in physics-based models. Drgoňa et al. (2020) and Smarra et al. (2018) show that tree-based models can also support predictive-control logic, which makes a move toward benchmark-based intelligent control methodologically more coherent than a sudden move to high-fidelity white-box control or reinforcement learning without the required state and actuator information.

Accordingly, the thesis is positioned as a control-oriented analytical framework in which target reconstruction and expected-demand modelling serve as enabling layers for intelligent cooling control. The study is located closest to the practical end of the control spectrum, it does not implement validated closed-loop MPC or reinforcement learning, but evaluates forecast-based and context-aware control logic through bounded actions, explicit control states, and operating-mode discovery under real telemetry constraints.

1.6. Forecast-based benchmarking, control-oriented evaluation and thesis positioning

Once predictive modelling and intelligent control logic have been established, the next methodological question is how their operational value should be evaluated. In building-energy research, this is typically addressed through baseline modelling and measurement-and-verification logic. Qaisar and Zhao (2022) describe the energy baseline as a reference estimate of how a building would be expected to consume energy under given conditions, while Alrobaie and Krarti (2022) show that predictive models become practically useful when they serve as defensible expected-demand baselines rather than as isolated forecasting tools. In the context of the present thesis, this means that both the forecast model and the proposed control strategies must be interpreted relative to historical operation and expected demand, not only through prediction error but through their implications for benchmark-based cooling assessment.

This benchmarking perspective is particularly important when cooling demand is reconstructed from incomplete or fragmented metering and when the available data do not provide direct thermal-comfort or indoor-state confirmation. Canale et al. (2024) define abnormal thermal consumption as a significant deviation from a reference value and show that such diagnosis remains possible under limited metering when benchmarking is performed systematically. Chiosa et al. (2022) likewise argue that anomaly detection depends on an appropriate context-sensitive baseline and on careful feature selection, rather than on raw consumption inspection alone. For the present thesis, these studies support a restrained interpretation of positive residuals and control-trigger hours. They are best understood as model-relative evidence of possible excess cooling demand, not as direct proof of physical overcooling.

A further issue concerns how reduction potential should be interpreted once forecast-based or supervisory control strategies have been simulated. The literature does not support translating all positive residuals or all controller activations directly into actionable savings. Alrobaie and Krarti (2022) note that M&V outcomes remain sensitive to modelling choices and application conditions, while Chiosa et al. (2022) show that anomalous behaviour should be prioritised in context rather than treated uniformly. Stoffel et al. (2023) strengthen this point from the control perspective by showing that advanced control strategies must be compared under common conditions and against credible baselines if their operational value is to be interpreted meaningfully. In the present thesis, this supports the use of bounded actions, persistence requirements, operating-mode filters, and comparison against historical operation as a structured way to evaluate intelligent control strategies rather than as a claim of directly achievable savings.

The specific research gap addressed by this thesis lies at the intersection of telecom cooling diagnostics, short-horizon cooling-demand forecasting, and intelligent control under fragmented operational telemetry. Existing telecom studies are strong in monitoring, KPI design, and diagnosis, but provide less direct support for reconstructing a usable cooling-demand target from distributed cumulative meters, forecasting that target at short hourly horizons, and translating forecast outputs into benchmark-based control logic. Forecasting studies, meanwhile, often assume ready-made targets and stop at predictive accuracy, whereas advanced-control studies usually assume richer state feedback, manipulated variables, and validated control environments than are available in the present telecom case. This thesis addresses that gap through an applied framework for a real telecom central office that combines target engineering, comparative rolling-origin forecasting, expected-versus-actual benchmarking, and historical evaluation of intelligent control strategies.

Finally, the literature supports the thesis at four linked levels. First, telecom central-office studies justify treating cooling as a benchmark-oriented operational problem rather than as generic HVAC background. Second, preprocessing, sub-metering and target-engineering literature supports the reconstruction of an hourly cooling-demand target from fragmented cumulative measurements. Third, forecasting literature supports comparative time-series modelling under strong autoregressive structure, weather sensitivity and horizon-dependent behaviour. Fourth, predictive-control and benchmark-oriented evaluation literature show that model outputs can be used not only for prediction, but also for control-oriented decision logic and structured historical comparison. The specific contribution of the present thesis lies in combining these strands within one real telecom-site case to support intelligent cooling control under realistic sensing constraints.

2. RESEARCH DESIGN, DATA AND METHODS

2.1. Research design and analytical workflow

This thesis adopts an applied quantitative research design based on retrospective analysis of historical monitoring telemetry from a selected telecom central office. The study is empirical and observational. It analyses previously recorded operational data rather than data generated through controlled experiments or live intervention. The aim is to develop and evaluate intelligent cooling control strategies under real operational constraints, using predictive modelling as a support layer for control.

Telecom central offices are continuously cooled environments in which thermal stability is required for the reliable operation of sensitive electronic equipment. At the same time, the available telemetry is fragmented, unevenly complete, and not structured as a ready-made analytical dataset. For this reason, the study is designed as a single-site case study that prioritises methodological depth, traceability, and operational realism over broad but shallow cross-site comparison.

The research design is structured around three linked analytical questions. The first asks to what extent intelligent cooling control strategies can identify opportunities to improve cooling energy use in telecom central office environments. The second asks which forecasting approach provides the most reliable estimate of expected cooling demand for intelligent cooling control. The third asks which operational and environmental variables are most relevant for constructing a control-oriented analytical basis for intelligent cooling control. These questions are not independent. The control question is central, while the forecasting and data-basis questions support it. The design therefore follows a sequential analytical logic.

This structure is consistent with the research gap identified in the literature review. The reviewed telecom literature is comparatively strong in monitoring, KPI development, and diagnosis, but weaker in the full chain from target reconstruction to benchmark-based control analysis. The contribution of the thesis is therefore not the proposal of a new control algorithm or a new forecasting architecture in isolation. Rather, it is the development of an integrated applied framework that combines target engineering, comparative forecasting, benchmark-based evaluation, and intelligent control assessment within one real telecom central office case.

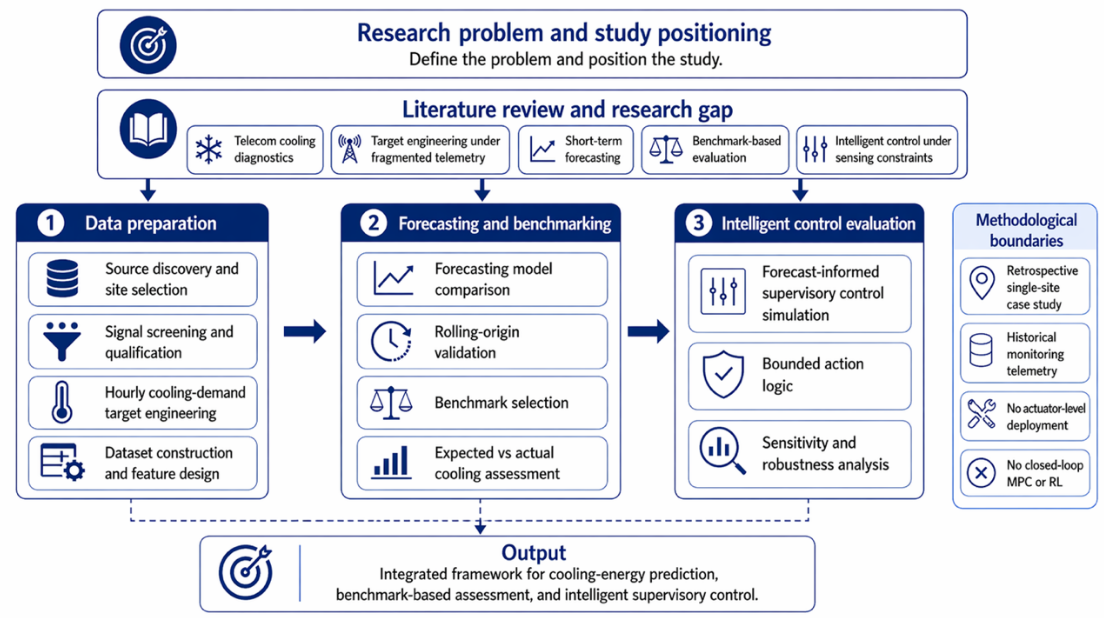

Fig. 1. Research design and analytical workflow of the thesis

Fig. 1 summarises the overall analytical workflow of the thesis. The workflow begins with source review and selection of a suitable telecom site. It then proceeds to signal screening and qualification, engineering of an hourly cooling-demand target from fragmented cumulative subsystem telemetry, dataset construction and feature design, and exploratory data analysis. On that basis, short-term forecasting models are developed and compared under chronological and rolling-origin validation in order to estimate expected cooling demand and select a defensible benchmark. The selected benchmark is then used for expected-versus-actual cooling assessment and for the evaluation of intelligent control strategies. The final stage consists of sensitivity and robustness analysis of the control design.

Although the workflow is primarily data-driven, it is not purely automatic. Engineering judgement enters at several points, particularly in site selection, signal qualification, target-component review, and interpretation of operating modes and control states. This is especially important because the cooling-demand target must be reconstructed before later benchmarking and control evaluation.

Within this design, predictive modelling serves two functions. First of all, to determine which model family provides the most reliable estimate of short-horizon expected cooling demand in the selected context. Secondly, to generate an expected-demand reference against which actual cooling behaviour can be interpreted. Forecasting accuracy is therefore important because the selected model becomes the benchmark for later assessment and control logic.

The intelligent control component of the thesis is implemented as a bounded and historically evaluated control framework. It is evaluated on historical operation rather than implemented as live actuator-level control. The study uses forecast deviations, contextual variables, explicit control states, operating modes, and bounded action rules to assess intelligent control strategies under the available telemetry conditions.

Overall, the research design can be described as a stage-gated analytical framework in which each phase produces the validated input for the next. The study moves from fragmented monitoring data to control-oriented operational insight through target engineering, expected-demand benchmarking, and intelligent control evaluation within one coherent telecom case.

2.2. Study site and data sources

2.2.1. Data access and extraction context

The empirical analysis was based on historical monitoring data obtained through read-only query access to a reporting database. Data extraction and preparation were carried out in Python. Because the accessible environment did not provide one ready-made modelling table, the workflow began with source discovery and metadata inspection rather than immediate bulk extraction of time-series values. Site-related tables provided contextual information for candidate locations, metadata tables described the available signals, measurement tables supplied the time-stamped values used later for profiling, target engineering, and dataset construction, and supporting utility-related records were used during early site screening. As a result, the empirical basis of the thesis was assembled progressively from multiple reporting sources rather than downloaded as a fixed analytical dataset.

2.2.2. Accessible source layers and analytical roles

The accessible reporting layer consisted of site-related, metadata, and measurement sources. These source layers were used for site identification, signal interpretation, cross-table mapping, measurement retrieval, target construction, and dataset assembly. A detailed summary of the source layers and their analytical roles is provided in Annex 2.

The site and support layer provided the structural and contextual basis for site selection and source interpretation. In particular, monitoring_site_model added model-level context for the selected site, including versioned site-model attributes and a structured technical site description through the doc field. In this thesis, that document is treated as a technical semantic reference used to interpret the site infrastructure, especially cooling-related subsystems and other equipment groups.

The signal metadata layer provided the descriptive basis for identifying and interpreting the available signal universe. It was central during the early stages of the workflow because it made it possible to screen candidate sites and later to define the set of signals relevant for detailed profiling and target construction. The measurement layer then provided the time-stamped values used for signal profiling, hourly aggregation, target engineering, and final dataset construction.

Taken together, these source layers show that the empirical basis of the thesis was assembled progressively from multiple reporting sources with different analytical roles. This staged structure is methodologically important because the later modelling and control-oriented analysis depended on prior source interpretation rather than on a ready-made dataset.

2.2.3. Selected study site and site-selection basis

The study site was selected through a staged screening procedure designed to identify one telecom central office with sufficiently rich cooling-relevant telemetry before heavy measurement extraction was attempted. The purpose of the screening was to identify one site with strong cooling-related signal richness, energy-related metadata support, at least minimal temperature or contextual support, and utility-related supporting information for downstream analysis.

In the first stage, enmos_datapoint was used as the primary source for site-level metadata screening. Textual fields were normalised and combined, and keyword groups related to cooling, energy, temperature, and status were used to assign signal-level relevance flags. These were then aggregated to site level to obtain counts of total signals and cooling-, energy-, temperature-, and status-related signals, together with a metadata-based site score.

In the second stage, site-level information from monitoring_site and monitoring_site_model was added to refine the ranking. The presence of a linked site-model record and path-level metadata relevance were combined with the earlier metadata score to produce a composite ranking value. Representative high-scoring signals from the top-ranked sites were then reviewed qualitatively to verify that the ranking was not dominated by irrelevant metadata noise.

In the third stage, the ranked sites were filtered to a reduced candidate set by requiring at least minimal energy and temperature support at metadata level. This step did not guarantee full modelling sufficiency, but it eliminated clearly unsuitable sites that lacked even basic evidence of energy and environmental-context signals.

In the fourth stage, the reduced candidate set was checked against supporting utility records in swenex_file_data. Coverage period, cost- and consumption-related fields, and general support availability were aggregated per site and combined with the earlier metadata score to form the final candidate ranking. Upper clipping was applied to prevent utility-related support terms from dominating the ranking.

The final selected site was the top-ranked site under this combined procedure. Relative to the second-ranked candidate, it showed a substantially richer signal environment at metadata level, including far more total, cooling-related, and energy-related signals, together with sufficient supporting information for later signal qualification and modelling. For confidentiality reasons, the site identifier is not disclosed in the thesis; throughout the text it is referred to simply as the selected telecom central office site.

This site-selection procedure is methodologically important because it makes the single-site design reproducible. The selected site was not chosen from prior familiarity or manual preference alone, but through an explicit screening procedure using the same keyword groups, scoring logic, candidate filtering rules, and utility-support enrichment steps. The result was a technically justified case-study basis for the later stages of signal qualification, target engineering, expected-demand modelling, and intelligent control evaluation.

2.2.4. Study period and rationale for the selected time window

The accessible historical data covered more than one calendar year, but the empirical analysis was restricted to the 2025 calendar year. This was a deliberate study-design choice rather than a database limitation. The selected window was intended to provide one recent, complete, and internally coherent annual period for downstream signal qualification, target engineering, expected-demand modelling, and benchmark-based analysis. At this stage, the choice was justified at the source level rather than through the final engineered cooling target, because target identification and reconstruction were performed later in the chapter.

A month-level continuity check of the selected site measurement layer was therefore used to compare the available annual windows. As shown in Fig. 2, the year 2025 provided the strongest basis for downstream analysis. Monthly temporal continuity remained effectively complete throughout the year, with only one moderate reduction in August, where continuity reached 94%. By contrast, 2024 showed visible disruption in spring, with continuity reduced to 74% in March and 20% in April, indicating that the annual record for that year was less stable for the planned workflow. The year 2026 was not retained as the main study period because, although the available early months showed high continuity, the year was still partial at the time of extraction.

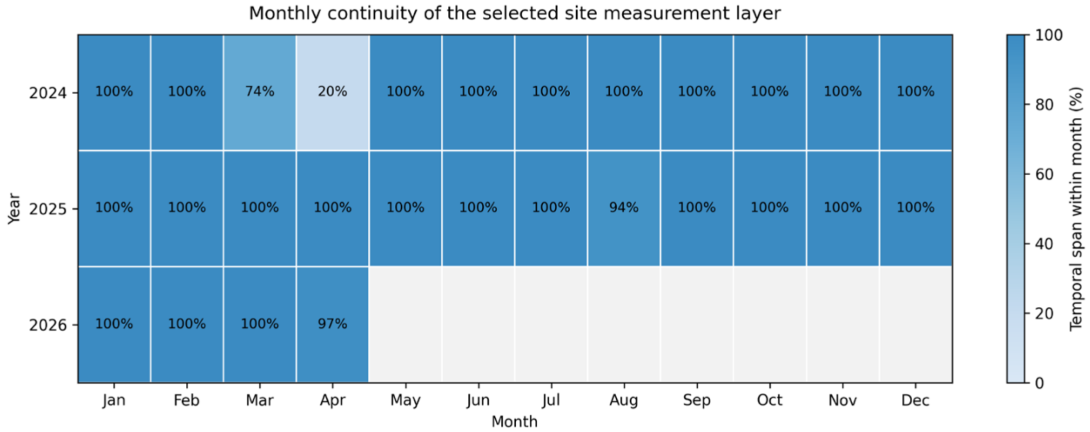

Fig. 2. Monthly temporal continuity of the selected site measurement layer across available years

The use of 2025 as the study period was methodologically important because the thesis required one sufficiently continuous annual basis for target reconstruction, seasonal comparison, chronological expected-demand modelling, and later benchmark-based control evaluation. On this basis, the study period was restricted to the 2025 calendar year. Later stages of the methodology then used this selected annual window for signal qualification, engineering of the hourly cooling-demand target, construction of the modelling dataset, comparative forecasting, and intelligent control evaluation.

2.3. Signal screening, cross-table mapping, and variable-selection logic

After establishing the database-access context and visible source structure, the next methodological step was to determine which site-specific signals could actually support downstream analysis. The visible metadata layer contained more candidate signals than could be assumed usable for target engineering, expected-demand modelling, or intelligent control evaluation. Signal screening was therefore treated as a staged qualification process in which metadata correspondence, measurement-layer availability, temporal continuity, and analytical relevance were assessed before any final target or predictor set was defined.

The screening workflow began with cross-table mapping between enmos_datapoint and datapoint for the selected site. In total, 567 ENMOS metadata rows and 223 datapoint rows were loaded. A direct mapping between datapoint.dp_id and ENMOS name showed that all 223 datapoint rows could be matched to ENMOS-side names, while 344 ENMOS rows remained unmatched. This indicated that the datapoint layer formed a complete and internally consistent subset of the broader

ENMOS-described site environment, whereas the ENMOS layer also contained additional metadata-visible rows that could not automatically be treated as technically usable measurement inputs.

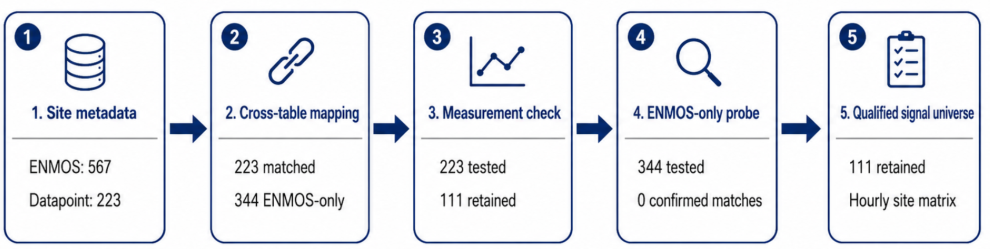

Fig. 3. Signal-screening and qualification workflow for the selected telecom site

As shown in Fig. 3, the key transition in this stage was from nominal metadata visibility to technically retrievable and measurement-supported signals suitable for later target engineering and expected-demand modelling. The 223 matched datapoint IDs were then profiled against the measurement and measurement_all tables in order to assess practical availability, temporal range, and basic value behaviour. This reduced the pool to 111 signals with confirmed measurement-layer support. A consistency check showed complete overlap between measurement and measurement_all for this retained set, confirming that the signals behaved consistently across both sources.

A separate availability probe was then conducted for the 344 ENMOS-side metadata rows that had no datapoint match. Because these rows lacked datapoint_id, they could not be queried through the normalized measurement table and were tested only through string-based lookup in measurement_all. Under the tested lookup strategy, no confirmed time-series matches were established. This did not imply that the ENMOS-only rows were meaningless, but it did mean that they could not be retained as directly retrievable signals in the same technically traceable way as the datapoint-mapped set. For that reason, the downstream analytical workflow remained anchored on the datapoint-mapped signal universe.

The retained datapoint-based signals were then transformed into a compact site-level inventory combining measurement profiling with datapoint metadata. The same set was aggregated to hourly resolution across the selected study period, producing a multivariate hourly signal matrix for the site. This qualified signal representation formed the empirical basis for the later stages of the thesis which are target-component review, target engineering, and predictor design.

The selection logic was deliberately sequential. A signal was retained only if it belonged to the selected site, was interpretable through metadata, was retrievable at the measurement layer, and contributed to a sufficiently continuous hourly representation of monitored operation. The output of this stage was therefore not a final feature set, but a technically qualified analytical signal universe from which later target and predictor selections could be made in a controlled and traceable way.

2.4. Engineering of the hourly cooling-demand target

A central methodological challenge of the thesis was that cooling demand was not available in the source environment as a directly usable hourly variable. Instead, the selected site contained a broader telemetry environment composed of subsystem measurements, cumulative energy signals, contextual variables, and partially overlapping metadata descriptions. For later expected-demand modelling and intelligent control evaluation, a defensible hourly cooling-demand target therefore had to be reconstructed from this fragmented signal environment.

This approach is consistent with the broader methodological position of the thesis. In monitored technical infrastructure, operational data are rarely analysis-ready, and sub-item energy behaviour often has to be inferred from distributed measurements rather than from one ready-made metered variable. For the present study, the cooling-demand target therefore had to be engineered explicitly before expected-demand modelling, benchmark-based assessment, or intelligent control evaluation could be justified.

2.4.1. Selection of cumulative subsystem signals

The initial candidate pool was intentionally broad and included cooling-family cumulative energy signals together with several linked signals retained for comparison. The full shortlist of candidate target components and their system roles is provided in Annex 3. The final target-component selection was based on cooling-family semantics, hourly usability, non-trivial variability, co-movement with related cooling signals, and interpretability within the site context.

The first step in target engineering was to identify which signals from the retained hourly signal universe could plausibly function as components of a site-level cooling-demand target. This selection was not performed mechanically from all cooling-labelled signals. Instead, it followed a staged review process combining cooling-family metadata semantics, hourly usability after aggregation, non-trivial variability, co-movement with other plausible cooling-family signals, and interpretability within the technical site context. The objective at this stage was not yet to define the final target, but to construct a technically plausible candidate pool for closer comparison.

The annual behaviour of the shortlisted candidates is illustrated in Fig. 4. through their monthly average profiles. The purpose of this comparison is to distinguish the most stable and coherent cooling-family subgroup from candidates that, although cooling-related, reflect more peripheral, branch-level, or subsystem-specific behaviour.

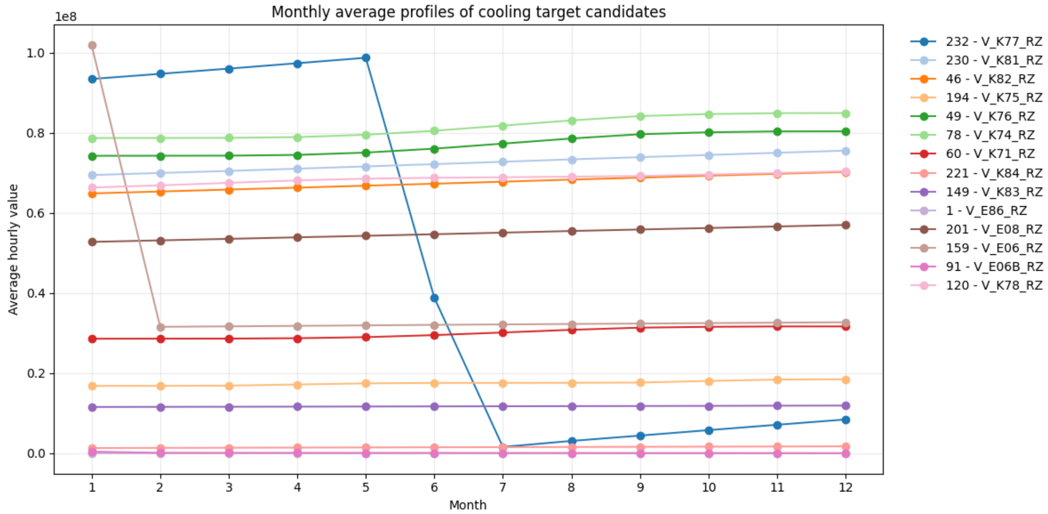

Fig. 4. Monthly average profiles of shortlisted cooling-target candidates

As shown in Fig. 4, several shortlisted candidates exhibit broadly stable annual behaviour, but they do not all represent the same technical role in the cooling system. Signals such as V_K71_RZ and V_K84_RZ remain cooling-related and structurally plausible, yet they correspond to narrower production-side or branch-level subsystem roles rather than to the core distributed cooling structure that the target was intended to represent. Similarly, V_E08_RZ, V_E06_RZ, and V_E06B_RZ showed strong linkage to the cooling environment but were interpreted more appropriately as infrastructure-adjacent or proxy signals than as components of the core cooling target.

On this basis, the final target components were restricted to datapoint IDs 120, 230, 46, 49, and 78, corresponding to V_K78_RZ, V_K81_RZ, V_K82_RZ, V_K76_RZ, and V_K74_RZ. These signals were retained because they jointly satisfied the full selection logic. They had clear cooling-family semantics, remained technically usable after hourly aggregation, showed non-trivial variability, formed a coherent co-moving signal group, and represented the most stable and interpretable core cooling-distribution subset among the shortlisted candidates.

2.4.2. Hourly alignment and differencing

After the final cumulative target components had been selected, the next step was to transform them into a common hourly representation suitable for interval-based target construction. The retained signals were already available in the site-level hourly signal matrix produced in the previous stage of the workflow. For target engineering, the five selected cumulative components were extracted from this matrix and aligned on a shared hourly timestamp index.

This step was necessary because the retained series represented cumulative subsystem measurements rather than direct hourly cooling demand. In cumulative form, the values describe running totals and therefore cannot be used directly as the target for later modelling or benchmark-based analysis. The aligned cumulative series were therefore converted into hourly increments using first-order differencing so that each hourly value represented the change from one timestamp to the next.

The differencing step changed the interpretation of the retained signals from cumulative subsystem totals to hourly interval contributions. At the same time, differencing cumulative telemetry introduces known practical issues. In real metering environments, negative steps may occur because of counter resets, irregular reporting, missing intervals, or other non-physical artefacts rather than true negative cooling consumption. For this reason, differencing was treated separately from the later cleaning and aggregation stage.

2.4.3. Cleaning, plausibility review, and aggregation into the final target

After hourly alignment and differencing, the retained component series could not yet be treated as a valid cooling-demand target without additional cleaning and review. In the implemented workflow, cleaning was first applied at component level. After differencing, each retained cumulative cooling component was clipped at a lower bound of zero. This reflects the interpretation adopted in the thesis. Negative hourly increments in cumulative cooling-energy counters were treated as non-physical artefacts rather than as valid negative cooling demand. The purpose of this step was therefore conservative cleaning rather than statistical smoothing.

Once the negative component-level artefacts had been removed, the cleaned hourly component contributions were summed to form the final target. The resulting series represents an engineered hourly cooling-demand variable reconstructed from fragmented cumulative subsystem telemetry rather than a directly metered ready-made load series.

After aggregation, the target was reviewed for plausibility before being carried forward into dataset construction.

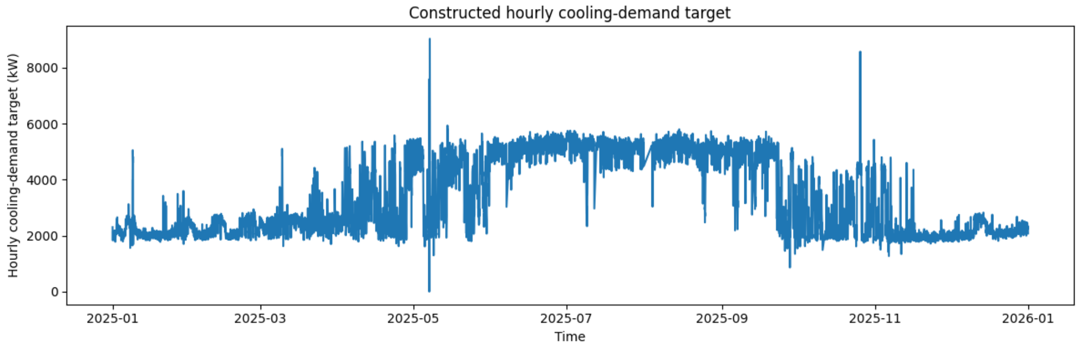

Fig. 5. Constructed hourly cooling-demand target for the selected study period

As shown in Fig. 5, the target exhibits a clear annual structure. Lower and comparatively stable levels are visible in winter, the series rises through spring, remains elevated and more variable during the warm season, and then declines again toward late autumn and year end. The figure also shows several isolated spikes and sharp dips, including a small number of extreme values, but these remain sparse relative to the broader annual pattern. The visual review therefore supports the interpretation that the engineered target forms a coherent hourly cooling-demand series suitable for later dataset construction and expected-demand modelling.

Following this plausibility review, the target was retained for downstream dataset construction with only light later treatment of extreme values rather than aggressive filtering. In methodological terms, the thesis therefore distinguishes between clearly implausible component-level negative steps, which were removed directly before aggregation, and higher-level target outliers, which were first reviewed diagnostically and then treated conservatively in later stages.

2.5. Dataset construction and feature design

After the hourly cooling-demand target had been engineered and lightly cleaned, the next step was to construct the modelling datasets used in the subsequent expected-demand modelling, benchmark-based evaluation, and intelligent control stages. This stage was not treated as a simple merge of all available signals. Instead, the predictor space was structured deliberately in order to preserve interpretability and to limit unnecessary proximity between predictors and the engineered target. Two related datasets were therefore assembled: a compact main dataset, intended as the primary analytical basis of the thesis, and a broader comparison dataset, used to test whether additional target-adjacent sibling cooling signals provided meaningful extra predictive value.

The dataset-construction logic followed four principles. First, the final target remained separate from the predictor space. Second, predictors were grouped into internal operational variables, external weather context, and calendar variables. Third, sparse internal meteo-like signals were reviewed explicitly rather than retained automatically. Fourth, sibling cooling signals were isolated in a comparison dataset rather than mixed directly into the main specification.

2.5.1. Main dataset

The main dataset combined the cleaned hourly cooling-demand target with three predictor groups: internal operational telemetry, external weather variables, and calendar or time features. In the final assembled version, the predictor block contained 36 internal operational variables, 5 retained weather variables, and 4 calendar variables, giving 45 predictors in total. After the target variable was added, the dataset contained 46 columns. Following final row-level cleaning, the retained main dataset contained 8603 hourly observations spanning the selected 2025 study period. This dataset was treated as the primary modelling basis of the thesis.

2.5.2. Comparison dataset

In parallel with the main dataset, a comparison dataset was assembled to test whether a broader cooling-related predictor space would materially improve forecasting performance. This extended version retained the same target, internal operational block, weather variables, and calendar features, but additionally included 23 sibling cooling-related signals that were more closely connected to the target family. Before row-level cleaning, this yielded 68 predictors plus the target; after final cleaning, the resulting dataset contained 8603 rows and 69 columns. This dataset was used as a controlled comparison variant, while the main dataset remained the preferred analytical basis for expected-demand modelling and later control-oriented interpretation.

2.5.3. External weather integration and justification

External weather data were integrated into the study because cooling demand at a telecom central office is influenced not only by internal operational behaviour but also by ambient atmospheric conditions. In the available site telemetry, two weather-related variables were present in the monitoring environment: outside temperature and outside radiation. However, these internally available meteorological signals did not provide a sufficiently complete hourly basis for modelling. Their temporal coverage was sparse, with only a small number of observations per day and substantial missingness across the study period. In practical terms, this made them unsuitable as the main environmental predictor block for expected-demand modelling.

For this reason, external weather data were retrieved for the selected site location and aligned to the same hourly timestamp basis as the engineered cooling-demand target and the retained internal operational variables. For confidentiality reasons, the exact site location is not disclosed in the thesis. The purpose of this step was to provide a stable hourly representation of the external atmospheric conditions under which cooling demand evolved.

The methodological question at this stage was whether the externally retrieved weather series could serve as a valid environmental context despite the limitations of the internally available meteorological records. To assess this, the available site-level outside temperature signal was compared with the external air-temperature series over the overlapping period.

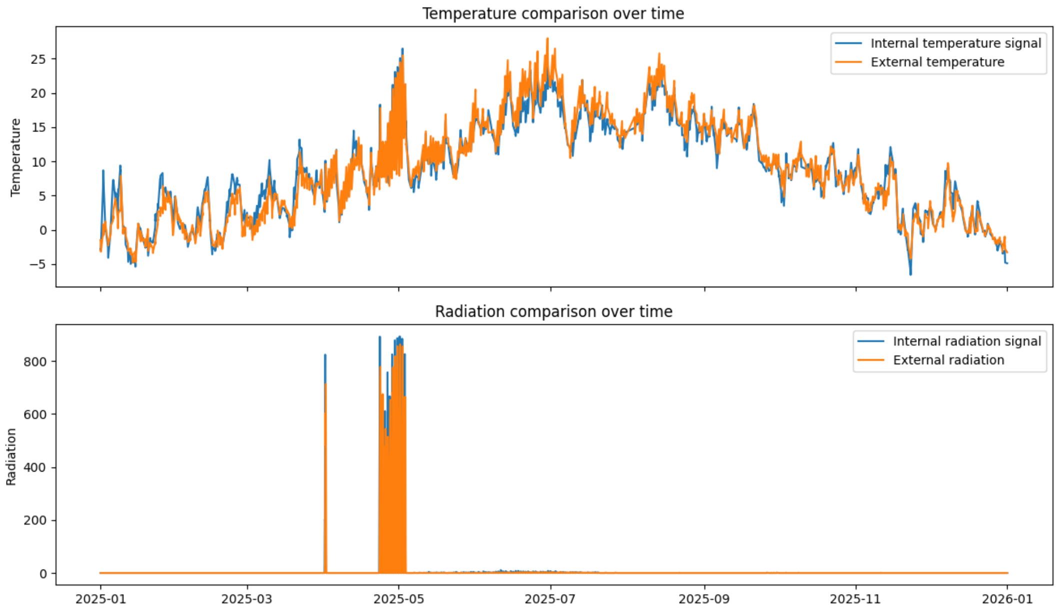

Fig. 6. Comparison of internal temperature, radiation proxy and external temperature context

As shown in Fig. 6, the broad temporal behaviour of the two series is coherent, even though the internally available outside-temperature record is much less complete and therefore not suitable as the main weather source for modelling. This comparison supports the use of external weather data as the primary atmospheric context in the final dataset design.

On this basis, external weather variables were retained as the main environmental predictor block for downstream modelling. In the final main dataset, five external weather variables were included as the core atmospheric context. At the same time, no direct indoor room-temperature measurements were available in the source environment, and the study therefore relies on external weather and internal operational telemetry rather than richer internal thermal-state feedback.

2.6. Exploratory data analysis

After dataset construction, an exploratory data analysis was performed to review the final engineered cooling-demand target and the main modelling. The purpose of this stage was to confirm that the target and predictor space showed interpretable temporal structure, acceptable dataset completeness, and visible relationships that could justify the later modelling design. In this sense, exploratory analysis served as a methodological checkpoint between dataset construction and expected-demand modelling.

Following final row-level cleaning, the retained main dataset contained 8603 hourly observations and 46 columns, while the comparison dataset contained the same 8603 hourly observations and 69 columns. In both datasets, missingness was negligible after final cleaning, so the exploratory analysis focused primarily on temporal structure and visible variable relationships rather than on data-repair issues. This was methodologically important because later modelling stages relied on chronological validation and lagged features, which are most interpretable when the retained dataset is temporally coherent and broadly complete.

The first exploratory focus was the annual structure of the engineered cooling-demand target. The full-year target series showed a clear seasonal pattern, with lower and comparatively stable levels during winter, rising demand through spring, elevated and more variable behaviour during the warm season, and a decline again toward late autumn. This broad pattern was already visible in the target-construction stage and remained present in the final modelling dataset. To examine this seasonal structure more systematically, the monthly distribution of the target was reviewed across the selected study period.

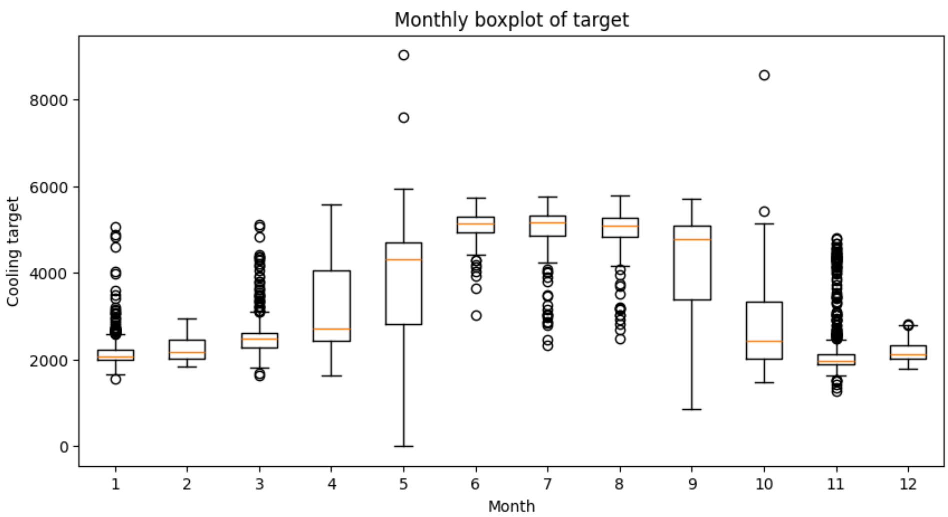

Fig. 7. Monthly distribution of the engineered hourly cooling-demand target

As shown in Fig. 7, the target distribution varies substantially across the year. Winter months are characterised by lower median levels and narrower dispersion, whereas spring and summer months show both higher demand and wider spread. This indicates that the cooling-demand target is not only seasonally structured, but also more operationally variable during warmer periods. The result is consistent with the interpretation that the annual window captures a meaningful cooling regime rather than a structurally flat site process.

The second exploratory focus was the within-day structure of the target across seasons. For this purpose, average hourly profiles were compared between winter, spring, summer, and autumn. This analysis showed that the cooling-demand target retains a visible time pattern within each season, while the level and shape of that profile change across the year. In practical terms, the target was therefore not only seasonally dependent, but also shaped by short-term temporal dynamics.

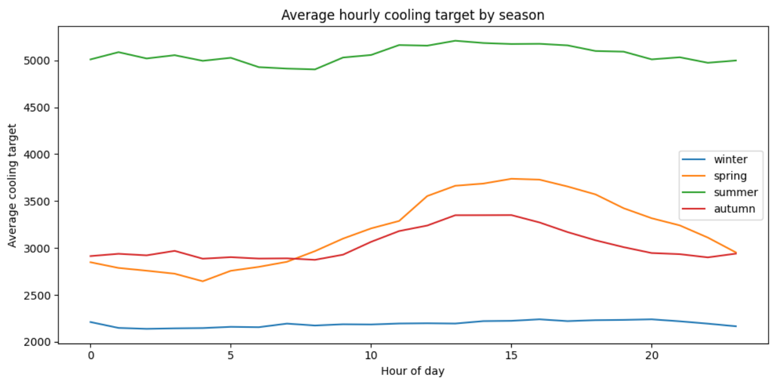

Fig. 8. Average hourly cooling-demand profile by season

Fig. 8 shows that seasonal differences in cooling demand are accompanied by visible within-day structure, supporting the use of lagged and context-aware modelling. Fig. 8. shows that the hourly structure of the target differs across seasons, with the warm season remaining consistently elevated across much of the day and the colder seasons operating at a lower overall level. This reinforces the interpretation that short-horizon cooling demand is shaped by both persistent recent history and broader seasonal context, rather than by one isolated external driver alone.

In addition to temporal structure, the exploratory analysis also reviewed the most visible relationships between cooling demand and selected weather variables. In the final main dataset, the target showed strong positive correlation with outside temperature and dew point, with Pearson correlations of approximately 0.851 and 0.813 respectively. Shortwave radiation also showed a moderate positive relationship, while relative humidity and cloud cover were negatively correlated with the target at approximately -0.413 and -0.259. These patterns indicate that cooling demand at the selected site is strongly weather-sensitive, but not in a purely one-dimensional way. Instead, the target reflects a combination of thermal exposure, humidity-related atmospheric conditions, and internal operational dynamics.

Taken together, the exploratory analysis supported three methodological conclusions for the later stages of the thesis. First, the engineered hourly target showed a coherent annual and seasonal structure suitable for expected-demand modelling. Second, the final datasets were sufficiently complete after cleaning to support chronological evaluation without major data-repair distortions. Third, the target displayed both strong short-term persistence and visible weather dependence, which justified the later use of lagged target features, weather variables, and comparative model families in the forecasting stage. The exploratory analysis therefore served not as a separate result chapter, but as a methodological bridge from dataset construction to formal expected-demand benchmark development.

2.7. Forecasting methodology and validation design

After the target engineering, dataset construction, and exploratory review stages had been completed, the next methodological task was to define how expected cooling demand would be modelled and how competing forecasting approaches would be evaluated. In the present thesis, forecasting is not treated as an isolated objective. Its purpose is to provide a reliable estimate of expected cooling demand that can later serve as the benchmark for expected-versus-actual assessment and for intelligent control evaluation. The forecasting stage therefore had to be designed not only for predictive accuracy, but also for methodological transparency, robustness, and suitability for downstream interpretation.

The forecasting design was built around a comparative approach. Rather than assuming in advance that one model family would be dominant, the study evaluated both simple benchmark formulations and more flexible machine-learning models under the same chronological conditions. This was important because the engineered cooling-demand target showed both strong short-term persistence and clear seasonal structure, which meant that simple baselines could already be competitive at short horizons, while more expressive models might still offer gains when lagged information, weather context, and nonlinear interactions were taken into account.

A further methodological requirement was that model evaluation should reflect the real temporal structure of the data. Because the study is based on hourly historical telemetry from one selected annual window, random train-test splitting would not have been appropriate. Instead, the forecasting design relied on chronological separation and rolling-origin evaluation, so that each model was assessed under repeated forward-looking conditions that more closely resemble operational use. This also ensured that model comparison remained consistent with the broader control-oriented logic of the thesis, where expected demand must be estimated from past and current information rather than inferred from randomly mixed observations.

The forecasting methodology was therefore organised around five components. First, the forecasting objective and analytical setup were defined. Second, benchmark and machine-learning model families were selected for comparative evaluation. Third, predictor groups and lag design were specified. Fourth, chronological split and rolling-origin validation were established as the main evaluation framework. Fifth, model performance was assessed using a small set of complementary error metrics and benchmark-selection criteria.

2.7.1. Forecasting objective and analytical setup

The forecasting task in this thesis was defined as short-horizon prediction of the engineered hourly cooling-demand target for the selected telecom central office site. The primary objective was one-hour-ahead forecasting. This horizon was chosen because it is sufficiently short to remain operationally relevant for benchmark-based assessment and intelligent control logic, while still allowing comparison between simple persistence-dominated baselines and more flexible model families. In the thesis, the forecast output is not treated as a final product in itself. Its methodological role is to provide an estimate of expected cooling demand against which actual cooling behaviour can later be interpreted.

The forecasting problem was formulated on the final hourly target constructed in Section 2.4 and on the modelling datasets defined in Section 2.5. The main dataset served as the primary analytical basis of the forecasting stage because it offered the strongest balance between predictive usefulness, interpretability, and lower target-proximity risk. The comparison dataset was retained as a methodological variant in order to test whether the inclusion of additional sibling cooling signals materially improved predictive performance. In this way, the forecasting setup allowed both benchmark construction and a controlled evaluation of whether broader target-adjacent information justified the loss of analytical clarity.

From an operational point of view, the forecasting task was designed as direct next-step estimation of hourly cooling demand from currently available and lagged information. The target variable was therefore the engineered hourly cooling-demand series, while the predictor space consisted of internal operational telemetry, external weather context, calendar features, and selected lagged values introduced later in this section. This setup reflects the intended role of the forecast within the thesis. To generate a short-horizon expected-demand reference that can support later benchmark-based interpretation.

The forecasting stage was also designed under the real-data constraints of the study. The available data cover one selected annual window rather than multiple long historical years, and the target itself is engineered from fragmented cumulative subsystem telemetry rather than directly metered as a native hourly load. For this reason, the modelling setup emphasised methodological transparency and stable comparison across model families. The objective was not to maximise apparent fit under arbitrary predictor expansion, but to identify a defensible expected-demand benchmark that could later be used for expected-versus-actual assessment and intelligent control evaluation.

Although the central modelling horizon was one hour ahead, the forecasting design was not intended to imply that one-hour prediction is the only meaningful horizon for cooling analytics. Rather, it was selected as the primary benchmark horizon because it aligns with the short-term operational logic of the thesis and with the strong persistence structure observed in the exploratory analysis. Broader horizon sensitivity was considered separately later as a methodological extension rather than as the main modelling objective. On this basis, the next step was to define which benchmark and machine-learning model families would be included in the comparative forecasting experiment.

2.7.2. Model families and benchmark candidates

The comparative forecasting experiment included both simple benchmark formulations and more flexible machine-learning models. This design was used to test whether more expressive methods offered a meaningful improvement over strong short-horizon baselines under the same data and validation conditions. The full list of evaluated models and their roles in the experiment is provided in Annex 4.

The first benchmark layer included naive persistence, seasonal naive forecasting, ARIMA, SARIMA, linear regression, random forest, and an initial LSTM configuration. This layer was used to establish the strength of simple and classical forecasting approaches before explicit lag engineering was introduced. In this setup, naive persistence was retained as an important reference because short-horizon cooling demand was expected to be strongly shaped by recent demand history.

The second benchmark layer introduced an explicit lagged forecasting setup. The main compared models were linear regression with lags, random forest with lags, gradient boosting with lags, and a multivariate LSTM. This layer tested whether engineered autoregressive structure could improve expected-demand estimation relative to the initial comparison. The lagged tree-based models were especially relevant because they could capture nonlinear relationships between recent demand, weather context, calendar structure, and operational variables without requiring a fully specified physical model.

Classical time-series models were also retained in an exogenous extension through ARIMAX and SARIMAX with a limited weather input set. This made it possible to compare machine-learning models with autoregressive time-series approaches that explicitly included external weather context. In the later rolling-origin comparison, the retained classical representative was SARIMAX with exogenous inputs.

Taken together, the final model set was not intended to represent every possible forecasting family. Its purpose was to provide a defensible comparison between transparent baselines, classical autoregressive models, nonlinear machine-learning candidates, and neural sequence models under a shared temporal validation design. This was sufficient for the aim of the thesis because the forecasting stage was used to select a reliable expected-demand benchmark for later expected-versus-actual assessment and intelligent control evaluation.

2.7.3. Predictor groups and lag design

The forecasting design combined three predictor groups with an explicit lag structure. The first group consisted of internal operational telemetry retained through the earlier screening and dataset-construction stages. The second group consisted of external weather variables, introduced to represent the atmospheric context under which cooling demand evolved. The third group consisted of calendar and time features used to capture recurring temporal structure. Together, these groups formed the non-lagged predictor basis of the main and comparison datasets. On top of this basis, a structured set of lagged target features was introduced in order to represent the strong autoregressive behaviour identified in the exploratory analysis and initial benchmark experiments.

The lag design was not chosen arbitrarily. The initial benchmark comparison showed that naive persistence was already a strong reference at one-hour-ahead horizon, which implied that recent target history contained essential predictive information. For this reason, the later forecasting setup explicitly incorporated lagged values of the engineered cooling-demand target. In the implemented workflow, the retained lag set consisted of 1, 2, 3, 6, 12, 24, 48, and 168 hours. This set was designed to capture several temporal scales simultaneously: immediate short-term persistence, within-day dependence, multi-day carryover, and weekly recurrence. In this sense, the lag structure was intended to provide a compact but operationally meaningful representation of the temporal memory of the cooling-demand process.

The role of lagged target history in the thesis is methodologically important. The engineered cooling-demand target was not treated as a process driven only by exogenous weather or static site variables. Instead, it was assumed that current demand depends strongly on its own recent history, while weather and internal operational variables provide contextual refinement around that autoregressive core. This design is also consistent with the cooling-load forecasting literature reviewed in Chapter 1, where short-horizon cooling demand is typically described as both persistence-driven and weather-sensitive. In the present experiment, the lagged specification therefore served as the main bridge between the exploratory evidence of strong temporal dependence and the later comparative model evaluation.

The same lag logic was applied consistently across the main non-neural forecasting models in order to keep the comparative design interpretable. This meant that Linear Regression, Random Forest, and Gradient Boosting were all evaluated on a shared lagged target basis rather than on unrelated feature spaces. As a result, differences in performance between these models could be interpreted primarily in terms of model form and flexibility rather than in terms of unequal temporal information. The lagged setup was therefore not only a predictive device, but also a design choice that improved the fairness of the comparison.

At the same time, lagged target design was not allowed to collapse the forecasting problem into a purely autoregressive exercise. The final model inputs still retained internal operational variables, weather context, and calendar structure. This was important because the expected-demand benchmark later had to support interpretation, not just short-horizon fit. A model driven only by lagged values might still forecast well in a narrow technical sense, but it would provide a weaker analytical basis for later benchmark-based assessment of actual versus expected cooling. The final predictor design therefore preserved a balance between strong autoregressive information and broader contextual variables.

In summary, the predictor design of the forecasting stage combined retained internal operational variables, external weather context, calendar structure, and a compact set of lagged target features. This feature space was intended to reflect the actual character of the engineered cooling-demand process: strongly shaped by recent history, but also influenced by environmental and operational context. On this basis, the next methodological step was to define how the models would be trained and evaluated under chronological and rolling-origin validation.

2.7.4. Chronological split and rolling-origin validation

The forecasting models were evaluated under a strictly time-ordered validation design. Because the study is based on hourly operational telemetry from one selected annual window, random train-test splitting would not have been appropriate. In the present thesis, expected-demand modelling is intended to support later benchmark-based assessment and intelligent control evaluation, so the validation framework had to preserve the real temporal order of the data.

For this reason, the forecasting experiment used two complementary validation stages. The first was a chronological train-test split, used to establish an initial out-of-sample comparison under a fixed temporal separation. The second was rolling-origin validation, used as the main comparative framework for model evaluation and final benchmark selection. Fig. 9 illustrates the temporal logic of this design.

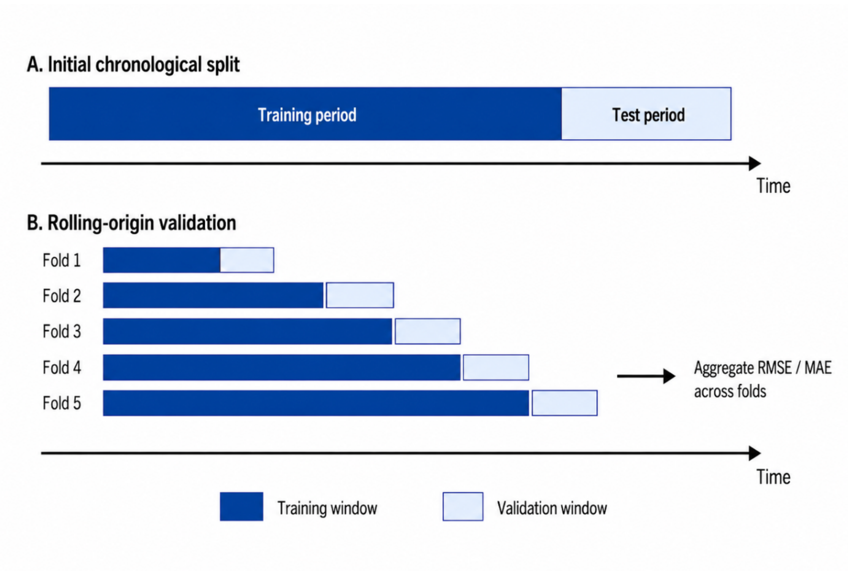

Fig. 9. Chronological split and rolling-origin validation design representation

In the chronological stage, the retained annual dataset was divided into an earlier training segment and a later testing segment without random reshuffling. This provides an initial out-of-sample estimate under realistic forecasting order, but one single split can still be sensitive to the specific time window chosen for testing.

For this reason, rolling-origin validation was used as the principal model-comparison framework. In this design, the forecast origin is moved forward repeatedly through time, and each model is re-evaluated on successive future validation windows. This makes the comparison less dependent on one particular holdout segment and better suited to a dataset with both short-term persistence and visible seasonal structure.

Rolling-origin validation was especially important because the later benchmark-based interpretation required a model that was not only accurate in one segment, but sufficiently stable to function as a defensible expected-demand reference. In the implemented workflow, rolling-origin validation was applied consistently across the retained benchmark, classical, and machine-learning candidates, and its results became the main input to the final benchmark-selection decision reported later in Chapter 3.

2.7.5. Evaluation metrics and benchmark selection

Model performance in the forecasting experiment was assessed using a small set of complementary evaluation metrics. The aim was not to maximise one isolated score, but to compare candidate models in a way that remained transparent and operationally meaningful for later benchmark-based use of the forecast. Because the selected model would later serve as the expected-demand reference, the evaluation framework had to reflect both average forecasting quality and robustness to larger deviations.

The two principal error metrics used throughout the comparative evaluation were mean absolute error (MAE) and root mean squared error (RMSE). MAE provides a direct measure of average absolute forecasting error in the original scale of the target, while RMSE is more sensitive to occasional large misses. Using both metrics therefore made it possible to compare the models from two complementary perspectives rather than relying on a single performance summary.

In addition to MAE and RMSE, the coefficient of determination (R^2) was reported for the final selected model as a descriptive measure of explained variance. However, R^2 was not treated as the main model-selection criterion. The central comparison and selection logic was based on direct forecast-error measures under chronological and rolling-origin evaluation, because these are more closely aligned with the practical role of the selected benchmark model.

Benchmark selection followed the same stage-gated logic as the broader methodology. Initial comparisons were used to identify plausible candidates and to verify whether persistence-dominated baselines already explained much of the short-horizon target behaviour. The main selection decision was then based on rolling-origin validation, where the retained models were compared repeatedly across unseen future segments. Under this design, the preferred benchmark model was the one that combined low forecast error with stable behaviour across folds rather than the one that performed best in one isolated holdout segment.

This selection logic was important because the benchmark model later becomes the reference for expected cooling demand and therefore affects the interpretation of residuals in the benchmark-based assessment stage. On this basis, the forecasting experiment used MAE and RMSE as the main comparative metrics, with R^2 reported as an additional descriptive indicator for the final selected benchmark.

2.8. Intelligent control methodology

After the expected-demand benchmark had been defined, the next methodological task was to translate that benchmark into a control-oriented analytical framework. In the present thesis, intelligent control is not implemented as actuator-level closed-loop control, full model predictive control, or reinforcement-learning control. Instead, it is implemented as a simplified sequential decision framework in which expected cooling demand, observed cooling demand, and contextual operating variables are combined to evaluate whether bounded intervention would have been plausible under historical operating conditions.

This positioning is consistent with the control literature reviewed in Chapter 1. Advanced cooling control is commonly framed within the broader family of predictive and constraint-aware control, where future demand, operating context, and action limitations are considered jointly. At the same time, practical deployment in technical infrastructure typically requires richer state feedback, actuator-level command variables, and validated dynamic models than were available in the present dataset (Drgoňa *et al.*, 2020; Killian and Kozek, 2016). For this reason, the thesis adopts a simplified control methodology that preserves the logic of predictive and context-aware decision-making while remaining consistent with the available telemetry.

From a conceptual perspective, the methodology can be interpreted using the core elements of sequential decision theory: state, policy, action, objective, and outcome. At each hourly step, the system is represented by a control-relevant state composed of forecast-based and contextual information. A rule-based policy then maps this state to a bounded action, and the resulting action is assessed retrospectively against historical operation. In this sense, the framework is related to the state-action logic used in sequential decision problems (Sutton and Barto, 2018), but its purpose here is practical rather than formal: to determine whether forecast-based and context-aware control logic can identify defensible intervention opportunities in historical cooling operation.

Within this framework, the state describes what the system knows at a given hour. In the present study, this includes expected cooling demand, actual cooling demand, the bias-corrected residual between them, seasonal eligibility, persistence information, a load proxy, and a thermal-context representation. The policy defines how this state is translated into a control decision. In the thesis, this is implemented through the strategy family B0-S3, where progressively richer policies incorporate not only residual information, but also persistence, load awareness, and operating-mode or risk-state awareness. The action space is intentionally simple and bounded: at each eligible hour, the methodology chooses either no action or a limited cooling reduction. The objective is to determine whether excess cooling can be reduced in a context-aware and operationally defensible manner, while avoiding aggressive or poorly justified intervention. The outcome is then assessed retrospectively through historical comparative evaluation, including reduction magnitude, action selectivity, and the distribution of actions across different operating-risk contexts.

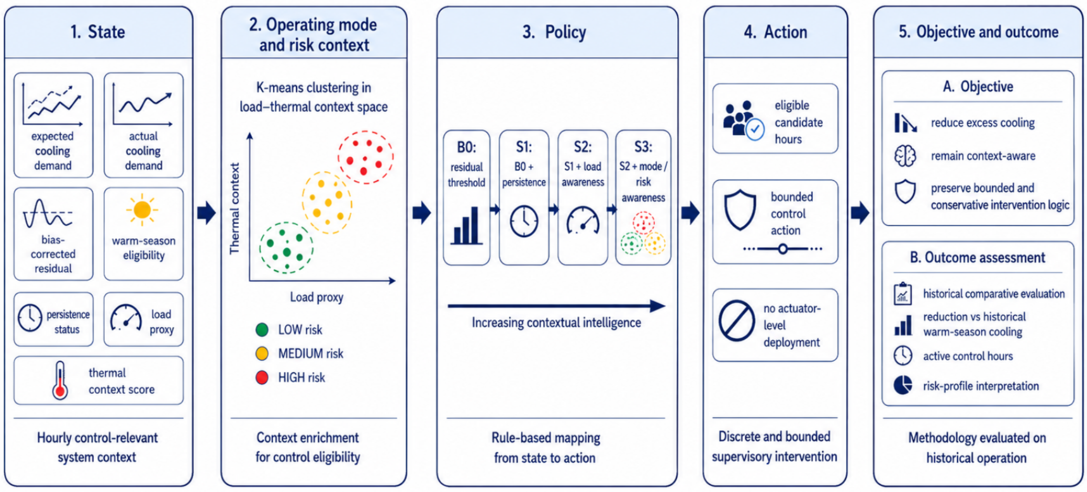

Fig. 10. Intelligent control methodology as a simplified sequential decision framework

Fig. 10 summarises the intelligent control methodology as a simplified sequential decision framework consisting of state construction, policy definition, bounded action, and historical outcome assessment. It shows that the methodology is conceptually related to state-action decision logic while remaining a historically evaluated and rule-based control framework. The following subsections describe each component of this framework in detail, beginning with the construction of the expected-versus-actual cooling logic that underlies the control state.

2.8.1. Control-state construction: expected-versus-actual logic and contextual variables

In the present thesis, the intelligent control methodology is built around an hourly control state. This state represents what the system knows at a given hour before any action decision is made. Rather than relying on one variable alone, it combines forecast-based, temporal, and contextual information into a compact control-relevant system description. Its purpose is to ensure that later control decisions are not triggered mechanically by observed cooling demand in isolation, but are interpreted relative to expectation, persistence, and operating context.

The state contains seven components: expected cooling demand, actual cooling demand, the bias-corrected residual between them, warm-season eligibility, persistence status, a load proxy, and a thermal-context score. Together, these variables define the control-relevant context from which the later policy rules determine whether bounded intervention is admissible. Expected and actual cooling define the benchmark comparison, the bias-corrected residual defines the main deviation signal, warm-season eligibility and persistence status define temporal admissibility, and the load proxy together with the thermal-context score provide the basis for operating-mode and risk-state interpretation.

The analytical core of the state is the comparison between expected and actual cooling demand. Expected cooling demand is obtained from the selected forecast model and treated as the benchmark estimate of what cooling demand would be expected under the observed conditions. Actual cooling demand is the engineered hourly target constructed earlier in the chapter from the retained cumulative subsystem signals. At each hourly timestamp, these two quantities are brought into direct comparison. This is methodologically central because the thesis does not interpret high cooling demand as problematic in itself, but asks whether the observed cooling level is elevated relative to a context-sensitive benchmark.

From this comparison, the deviation signal is constructed in the form of a bias-corrected residual. Positive residual values indicate that actual cooling remains above the benchmark after correction for small systematic model bias, while negative or near-zero values indicate that observed cooling is broadly consistent with expectation. In the present methodology, this residual is the main signal from which potential control relevance begins. However, it is not used alone, because positive deviation at one hour does not automatically imply a meaningful or admissible intervention opportunity.

For that reason, the control state also includes temporal admissibility variables. Warm-season eligibility restricts the methodology to the part of the annual cycle where cooling demand is structurally most relevant and where intervention logic is most interpretable. Persistence status distinguishes sustained excess from one-off fluctuation. Together, these two components reduce the risk of treating isolated or seasonally ambiguous deviations as meaningful control opportunities.

The final two state components provide contextual enrichment beyond residual-based eligibility alone. The load proxy represents the operational intensity of the site and allows the methodology to distinguish excess cooling under lighter and heavier system conditions. The thermal-context score represents the broader environmental context of the hour and is later combined with the load proxy to define operating modes and risk states. These variables are important because the same positive residual may have different practical meaning under different operating conditions. By including them in the control state, the methodology moves beyond simple thresholding toward a more context-aware interpretation of intervention admissibility. A fuller comparison of candidate load proxies is provided in Annex 5.

The control-state construction described in this subsection serves as the analytical foundation of the intelligent control methodology. It defines what information is available at each decision step and how that information is structured before any action rule is applied. The next subsection explains how the residual, seasonal, and persistence components of this state are used to filter the hourly timeline into a narrower set of candidate control hours.

2.8.2. Eligibility filtering: residual-based candidate control hours

After the hourly control state had been constructed, the next methodological step was to determine which hours were eligible for further policy evaluation. In the present thesis, this was done through residual-based eligibility filtering. The purpose of this stage was not to interpret every positive deviation from expected demand as a control opportunity, but to isolate only those hours in which excess cooling appeared sufficiently sustained and contextually plausible to justify later intervention logic. This distinction is important because short-term forecast deviation may arise from noise, transient operational fluctuation, or local model mismatch rather than from a meaningful control opportunity.

The starting point of this stage was the bias-corrected residual between actual and expected cooling demand. This residual served as the primary deviation signal because it represents observed cooling behaviour relative to a forecast-based benchmark while reducing the influence of small systematic model bias. Positive residual values indicate that actual cooling remains above the benchmark after correction, whereas negative or near-zero values indicate that observed cooling is broadly consistent with expectation. However, the residual alone was not treated as sufficient evidence for intervention, because one isolated positive hour may still be analytically weak or operationally irrelevant.

The first filtering layer was seasonal eligibility. In the implemented methodology, candidate control hours were restricted to the warm season. This restriction reflects the operational interpretation of the problem: the control logic was designed to identify potentially excessive cooling in the part of the year where cooling demand is structurally relevant and where bounded intervention can be interpreted more plausibly. Outside this seasonal window, positive residuals were considered less informative for control-oriented interpretation because the cooling regime is weaker and the meaning of intervention becomes less stable. Warm-season restriction therefore functioned as the first contextual boundary on the deviation signal.

The second filtering layer was persistence. In the present study, one isolated positive residual hour was not considered sufficient evidence of a meaningful control opportunity. Instead, the methodology required sustained excess before an hour could remain eligible for later strategy logic. This reflects the assumption that bounded intervention should respond to repeated or persistent deviation rather than to one-off anomalies. Persistence filtering therefore serves two methodological purposes: it reduces the influence of short-lived noise in the residual signal, and it aligns the control logic with a more cautious operational interpretation in which intervention is justified only when elevated cooling is not merely momentary.

Taken together, the warm-season and persistence filters transform the bias-corrected residual from a raw deviation measure into a narrower set of candidate control hours. This step is central to the methodology because it defines the transition from expected-versus-actual comparison to genuine control eligibility. In sequential-decision terms, the residual alone does not yet define an admissible action state. Only after seasonal and persistence conditions are satisfied does the hourly state become a plausible candidate for further policy evaluation.

This filtering logic is also consistent with the broader control literature reviewed earlier in the thesis. Simplified intelligent control in real infrastructure should not react mechanically to every deviation signal, but should incorporate temporal stability and contextual admissibility before bounded action is considered (Stoffel *et al.*, 2023; Maddalena, Lian and Jones, 2020). In the present study, residual-based filtering therefore functions as the first safeguard against overinterpreting model error as control opportunity.

The output of this stage is not yet a final action decision. It is a reduced set of eligible candidate hours that satisfy the minimum residual-based conditions for further control evaluation. The next subsection explains how these candidate hours are mapped through the strategy hierarchy B0-S3, where progressively richer policies add residual thresholding, persistence, load awareness, and operating-mode or risk-state awareness.

2.8.3. Policy hierarchy and bounded action design: strategies B0-S3

Once the candidate control hours had been defined through residual-based eligibility filtering, the next methodological step was to determine how intervention should be decided within that eligible state space. In the present thesis, this was implemented through a hierarchy of rule-based control policies denoted B0, S1, S2, and S3. These strategies are not independent controllers in a deployment sense, but progressively richer policy definitions that map the hourly control state to a bounded action. Their purpose is to test whether additional contextual intelligence improves the selectivity and interpretability of the control logic relative to simpler baseline formulations.

The strategy family is ordered by increasing contextual complexity. B0 is the simplest policy and functions as the methodological baseline. It applies a residual-threshold logic without richer contextual refinement beyond the minimum state construction already defined earlier. In this sense, B0 asks the most basic question: whether the observed excess relative to expected demand is large enough to justify a candidate intervention under the simplified control framework. Its role in the thesis is important because it provides the lowest-complexity reference against which the later strategies can be compared.

S1 extends this baseline by incorporating persistence explicitly into the policy logic. Under this strategy, intervention is not triggered by residual magnitude alone, but only when the excess signal also remains sufficiently sustained over time. The introduction of persistence makes S1 more selective than B0 and aligns it more closely with the practical idea that bounded intervention should respond to stable excess rather than to one-off deviation. Methodologically, S1 marks the transition from simple threshold logic to a genuinely time-aware control policy.

S2 adds load awareness to this persistent excess logic. Under this strategy, the intervention decision depends not only on the presence of sustained excess, but also on whether the operational intensity of the site remains compatible with bounded reduction. The use of a load proxy is important because the same residual does not necessarily imply the same control meaning under different operating conditions. A positive deviation observed under lighter operational load may be more interpretable as excess-cooling opportunity than a similar deviation observed under heavier or more constrained system conditions. S2 therefore introduces a clearer form of context-aware admissibility into the policy.

S3 is the richest policy in the implemented strategy family. It retains the residual, persistence, and load conditions of the earlier strategies, but also incorporates operating-mode or risk-state information. In this strategy, intervention is only allowed when the hourly state is located in an operational context judged sufficiently compatible with bounded action. This makes S3 the most selective and most context-aware strategy in the methodology. In sequential-decision terms, it is the policy that makes the fullest use of the available state representation, because it maps not only deviation and persistence, but also broader operating context, to the final action choice.

table 1.

Summary of the intelligent control strategy family B0-S3

Strategy	**Core decision basis**	**Added context**	**Intended role**
B0	Residual-threshold logic applied to the eligible state	None beyond basic eligibility conditions	Simplest baseline policy
S1	Residual-threshold logic with persistence requirement	Temporal stability of excess cooling	Time-aware policy
S2	Residual and persistence logic with load-proxy condition	Operational intensity of the site	Load-aware policy
S3	Residual, persistence, and load logic with operating-mode / risk-state condition	Broader operating context and risk compatibility	Most context-aware policy

As shown in Table 1, the strategy family is ordered by increasing contextual richness, allowing the methodology to compare simple and more selective policy formulations under the same bounded action framework.

Across all four strategies, the action space remains intentionally simple and bounded. The methodology does not attempt continuous control optimization or direct manipulation of equipment-level setpoints. Instead, each strategy maps the eligible hourly state either to no action or to a limited cooling reduction. The action is therefore discrete in decision form and bounded in magnitude. This design is methodologically important because it keeps the control framework aligned with the available telemetry and avoids overstating what could be justified from the historical data alone.

The strategy family B0-S3 can therefore be interpreted as a policy hierarchy of increasing contextual intelligence. B0 represents minimal residual-based action logic, S1 adds sustained excess, S2 adds operational-load context, and S3 adds operating-mode or risk awareness. This hierarchy is central to the thesis because it allows the control question to be examined comparatively: not only whether intervention opportunities can be identified, but whether richer and more context-aware policies lead to more selective and analytically defensible control behaviour.

This policy design is also consistent with the broader control literature reviewed earlier in the thesis. Simplified intelligent control in real infrastructure often takes the form of rule-based or approximate predictive control, where richer context and stronger admissibility logic are added progressively rather than solved through full online optimization (Stoffel *et al.*, 2023; Yang *et al.*, 2021; Maddalena, Lian and Jones, 2020). In the present study, the B0-S3 hierarchy serves exactly this role: it provides a structured comparison between simpler and more context-aware policies under the same historical conditions and bounded action logic.

2.8.4. Operating modes and risk-state definition

The richest policy in the intelligent control methodology required a more structured representation of operating context than residual, persistence, and load conditions alone. For this reason, an additional context layer was introduced in the form of operating modes and risk states. The purpose of this step was to distinguish hours that may show similar excess-cooling signals in magnitude, but occur under meaningfully different load-thermal conditions. In methodological terms, this allowed the control framework to move beyond threshold-based eligibility and toward a more context-aware interpretation of when bounded intervention was or was not plausible.

The operating-mode design was based on two state components already introduced earlier in the chapter: the selected load proxy and the thermal-context score. The load proxy represented the operational intensity of the site, while the thermal-context score represented the broader external thermal environment under which cooling demand evolved. Together, these variables formed a compact two-dimensional context space in which the hourly operating states of the site could be compared. This design was intentionally parsimonious, with the aim of capturing the main operational and environmental dimensions most relevant for bounded control interpretation.

To identify recurring operating situations within this context space, K-means clustering was applied to the hourly observations. K-means was selected because it provides a simple and interpretable partition of multivariate data into internally coherent groups, which is appropriate when the goal is operational categorisation of hours into a small number of control-relevant modes. This is consistent with the broader literature on data-driven operating-mode discovery, where clustering is often used to support supervisory interpretation under incomplete state feedback (Arbelaitz *et al.*, 2013; Rousseeuw, 1987; Davies and Bouldin, 1979).

The number of clusters was not fixed arbitrarily. Candidate partitions were reviewed using internal validity criteria together with practical interpretability. In particular, silhouette width and the Davies-Bouldin index were used as supporting numerical checks, while the final choice also required that the resulting groups remain meaningful when interpreted in load-thermal terms. In the present study, this combined review led to the retention of a three-mode structure.

The resulting operating modes were then interpreted as risk states for bounded intervention. This step converts unsupervised context clusters into a control-oriented admissibility layer. Hours located in the lowest-risk operating mode were interpreted as the most compatible with bounded intervention, because they combined more favourable load-thermal conditions with lower risk of unintended undercooling or operational stress. Hours in the middle mode were treated as intermediate or conditionally admissible. Hours in the highest-risk mode were interpreted as the least compatible with intervention and therefore the most constrained from the perspective of control logic. In this way, the operating-mode layer did not replace the earlier residual-based eligibility conditions, but enriched them by distinguishing whether a candidate hour occurred in a more or less intervention-compatible context.

This mode-to-risk interpretation is methodologically important because it makes S3 qualitatively different from the simpler strategies. B0, S1, and S2 decide primarily from residual behaviour, persistence, and operational load. S3, by contrast, adds a context-compatibility condition derived from the broader load-thermal state of the site. It therefore uses the fullest state representation available in the thesis and provides the most selective control policy in the strategy hierarchy.

As shown in Fig. 11, the clustering step partitions the hourly operating context into a small number of interpretable mode regions rather than a continuous undifferentiated state space. This makes it possible to incorporate operating compatibility into the control policy without requiring richer physical-state observability than the data can support.

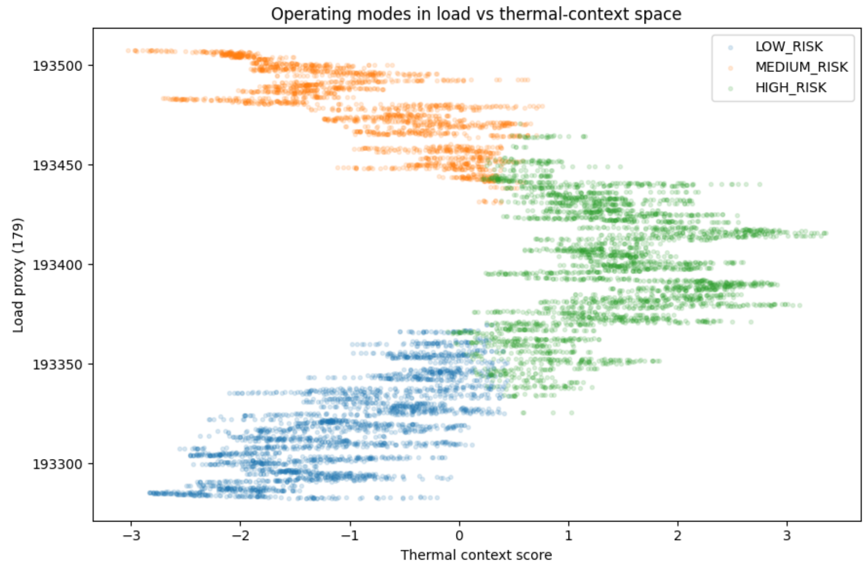

Fig. 11. Operating modes in load-thermal context space, interpretation as risk states

On this basis, the operating-mode layer became the final contextual component of the intelligent control methodology. The next subsection explains how the robustness of the full design was assessed through sensitivity analysis of the main control parameters and modelling choices.

2.8.5. Sensitivity-analysis design

Because the intelligent control methodology is built from several design choices rather than from one closed-form optimized controller, its robustness had to be examined explicitly before the later results could be interpreted. For this reason, the final stage of the methodology was a structured sensitivity analysis. The purpose of this stage was not to re-optimize the control framework exhaustively, but to test whether the main conclusions depended strongly on one narrow parameter choice or remained broadly stable across plausible alternatives. In the context of the thesis, this step was methodologically important because the proposed control logic is intended as a defensible analytical framework under real telemetry constraints rather than as a uniquely optimal control law.

The sensitivity design focused on the main choices that directly shape the selectivity and action intensity of the control framework. These included the residual threshold used to define meaningful excess, the persistence requirement used to distinguish sustained deviation from one-off fluctuation, the bounded reduction cap applied once a candidate hour was judged eligible, the choice of load proxy used in context-aware strategies, and the number of operating modes retained in the clustering-based risk-state layer. Together, these elements determine how strict or permissive the control logic becomes. Testing them explicitly was therefore necessary in order to assess whether the strategy comparisons remained interpretable when the control design was made somewhat stricter or somewhat more permissive.

In the implemented workflow, the residual threshold was varied through alternative quantile definitions of excess cooling. Rather than fixing one universal absolute threshold, the methodology defined meaningful excess relative to the distribution of positive residuals and then examined how the control results changed when this cut-off became more or less selective. Similarly, the persistence requirement was varied between shorter and stricter sustained-excess definitions in order to test whether the identified candidate hours depended strongly on one particular temporal rule. The bounded action layer was also subjected to sensitivity analysis by comparing alternative reduction caps, allowing the later evaluation to distinguish between the effect of candidate-hour selection and the effect of intervention magnitude.

A second part of the sensitivity design focused on contextual structure rather than thresholding alone. In particular, the methodology tested whether the control logic remained stable when different load-proxy formulations were considered and when the operating-mode layer was defined with alternative cluster counts. This was important because the richest policy in the framework, S3, depends not only on residual behaviour but also on the quality of the contextual partition used to interpret intervention risk. Sensitivity analysis at this stage therefore served as a safeguard against overinterpreting one arbitrary clustering choice or one narrow proxy definition as if it were uniquely correct.

The sensitivity design addressed the main assumptions shaping excess definition, temporal admissibility, contextual structure, and bounded intervention logic. The detailed sensitivity matrix is provided in Annex 6. The thesis did not attempt a full hyperparameter search over all possible policy definitions, but tested whether the main control conclusions remained stable when the most important methodological assumptions were varied within plausible ranges.

This robustness layer completes the methodological framework of the thesis and ensures that the later control results are interpreted against plausible alternative specifications rather than one fixed parameter setting alone. With the research design, expected-demand benchmark methodology, and intelligent control methodology now established, the thesis proceeds to the empirical results and analysis of the study.

3. RESULTS AND ANALYSIS

This chapter presents the empirical results of the thesis and interprets them in relation to the control-oriented aim of the study. The results are organised as a benchmark-based intelligent control analysis rather than as two separate forecasting and control stories. The chapter therefore begins with the selection of the expected-demand benchmark, because the forecast model must first be shown to provide a credible reference for later expected-versus-actual interpretation. It then proceeds to benchmark-based cooling assessment, the identification of candidate control hours, the comparative evaluation of the B0-S3 strategy family, the analysis of operating modes and risk states, and the robustness of the proposed control framework under alternative design choices.

This structure is consistent with the methodological logic established in Chapter 2. The expected-demand benchmark is not presented as an end in itself, but as the enabling layer for intelligent control analysis. For that reason, the benchmark results are reported only to the extent necessary to justify their later use in control-oriented interpretation. The main analytical emphasis of the chapter remains on whether benchmark-based and context-aware control logic can identify plausible opportunities to improve cooling energy use under real telemetry constraints.

3.1. Expected-demand benchmark selection

The first result question of the study was which forecasting approach provides the most reliable estimate of expected cooling demand for later control-oriented use. This question was addressed through a comparative forecasting experiment in which benchmark, classical time-series, and machine-learning models were evaluated under the same chronological and rolling-origin validation design described in Chapter 2. The purpose of this stage was not to identify the most complex model in abstract terms, but to select a forecast benchmark that was sufficiently accurate, stable, and interpretable to support later expected-versus-actual cooling assessment.

The results showed that the forecasting problem was strongly persistence-driven at one-hour-ahead horizon. Naive prediction provided a strong baseline, which confirmed that recent cooling-demand history contained substantial short-term predictive information. At the same time, persistence alone was not the upper limit of performance. The strongest results were achieved by lagged tree-based models, which outperformed both simpler baselines and the retained classical exogenous time-series alternative.

table 2.

Models performance results

Model	Rolling-origin MAE	Rolling-origin RMSE	Selection role
Naive	254.359444	382.450216	Reference baseline
Linear Regression + lags	287.317139	391.306857	Comparison candidate
Random Forest + lags	222.487411	319.491879	Final comparison candidate
Gradient Boosting + lags	228.051749	316.644365	Selected expected-demand benchmark
SARIMAX + exog	604.428176	829.977006	Classical reference candidate

As shown in Table 2, the strongest performance was concentrated in the lagged tree-based models. Gradient Boosting + lags achieved the lowest pooled RMSE, while Random Forest + lags achieved the lowest pooled MAE. Both clearly outperformed the Naive baseline, Linear Regression + lags, and SARIMAX + exog, whereas the classical exogenous time-series benchmark remained substantially weaker under the rolling-origin design. These results indicate that the expected-demand problem at the selected site benefits from nonlinear lag-aware modelling rather than from persistence or classical autoregressive structure alone.

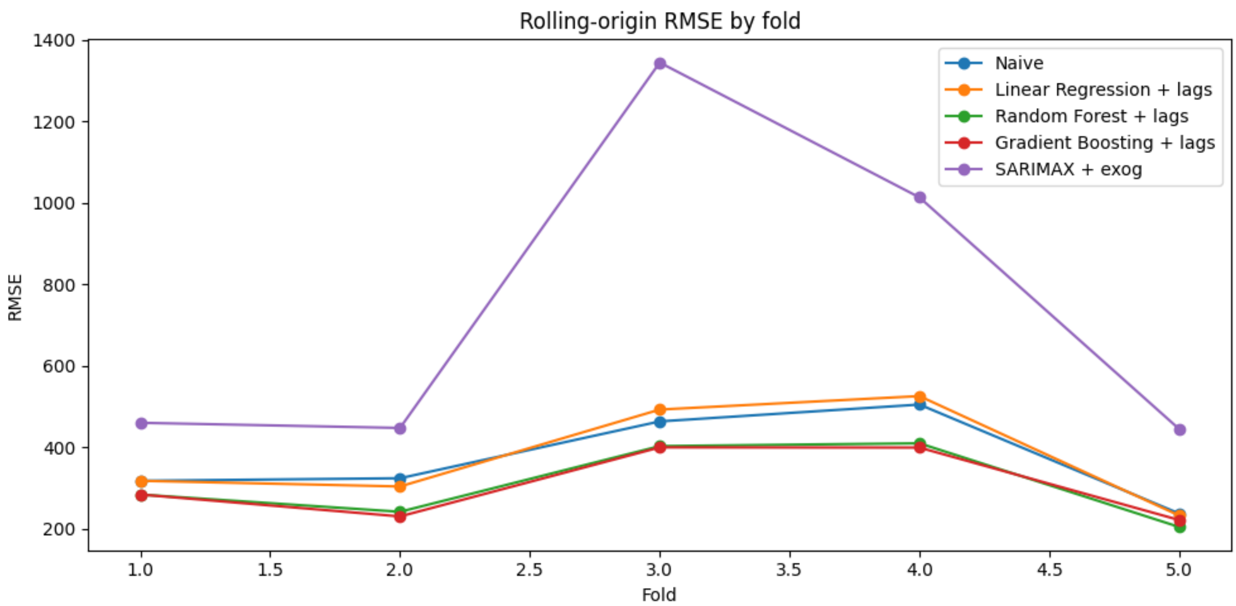

Fig. 12. Rolling-origin RMSE by fold for the main benchmark candidates

Fig. 12 shows that the relative ranking of the models was not driven by one single fold alone. The lagged tree-based models remained consistently strong across the repeated forward-looking validation windows, while SARIMAX + exog was visibly unstable and substantially weaker in several folds. The Naive and Linear Regression + lags models remained competitive in some windows, which further confirms that the target is strongly persistence-driven, but they did not match the overall rolling-origin performance of the best nonlinear candidates.

A brief interpretation of the selected benchmark also supports its analytical credibility. Feature-importance analysis of the retained Gradient Boosting model showed that lagged target history contributed most strongly to prediction, while external weather variables, especially temperature- and humidity-related features, provided additional explanatory value. This pattern is consistent with both the exploratory analysis and the literature review, which indicated that short-horizon cooling demand at the selected site is strongly persistence-driven but also materially shaped by weather context. The selected benchmark can therefore be interpreted not only as the best-performing model in the final comparison, but also as a plausible expected-demand representation for later control-oriented assessment. Extended feature-importance outputs for the selected benchmark model are provided in Annex 7.

The final benchmark-selection decision was based on repeated performance under rolling-origin validation and on suitability for later benchmark-based cooling interpretation. On this basis, Gradient Boosting + lags was retained as the final expected-demand benchmark model for the later stages of the thesis, with Random Forest + lags remaining the strongest alternative candidate. Once the expected-demand model had been selected, the next step was to examine how well it described observed cooling behaviour and how that benchmark could be used for benchmark-based cooling assessment.

3.2. Benchmark-based cooling assessment

After the expected-demand benchmark had been selected, the next analytical task was to examine how this benchmark changed the interpretation of observed cooling behaviour at the selected telecom central office site. In the present thesis, the forecast is not used only as a predictive output. Its main analytical role is to provide a context-sensitive reference against which actual cooling demand can be assessed. For this reason, the first control-relevant use of the selected Gradient Boosting benchmark is the comparison between expected and observed cooling demand.

This assessment was performed by comparing the engineered hourly cooling-demand target with the hourly expected-demand series produced by the selected benchmark model. The comparison allows observed cooling demand to be interpreted relative to what would be expected under the same historical operating, weather, and temporal conditions. Positive deviations indicate that observed cooling was above the expected level, while negative deviations indicate that observed cooling was below it. In this sense, the benchmark transforms the interpretation of site behaviour from absolute consumption alone to model-relative deviation. In the present thesis, this comparison is interpreted as evidence of potential excess cooling relative to the selected benchmark rather than as direct proof of physical overcooling.

The broad annual comparison did not indicate large-scale systematic benchmark-relative overcooling at site level. For the selected Gradient Boosting benchmark, the warm-season excess share was 1.3407%, the reduced share was 1.3610%, and the net balance was -0.0203%. A closely aligned result was obtained with Random Forest, which supports the conclusion that the aggregate annual interpretation is not driven by one single model specification. Taken together, these results suggest that the selected site did not exhibit strong evidence of persistent benchmark-relative overcooling at full-year scale. Instead, positive and negative deviations remained broadly balanced in the aggregate interpretation.

This annual balance does not imply that control-relevant deviation was absent. The comparison showed that positive and negative deviations were both present across the study period, and their operational meaning varied across time. In particular, the warm season remained the most relevant period for later control-oriented interpretation, because it combined the highest structural cooling demand with the clearest practical plausibility for bounded intervention. The later control logic does not rely on the full-year balance alone, but moves from aggregate benchmark interpretation to the identification of more selective benchmark-relative candidate hours.

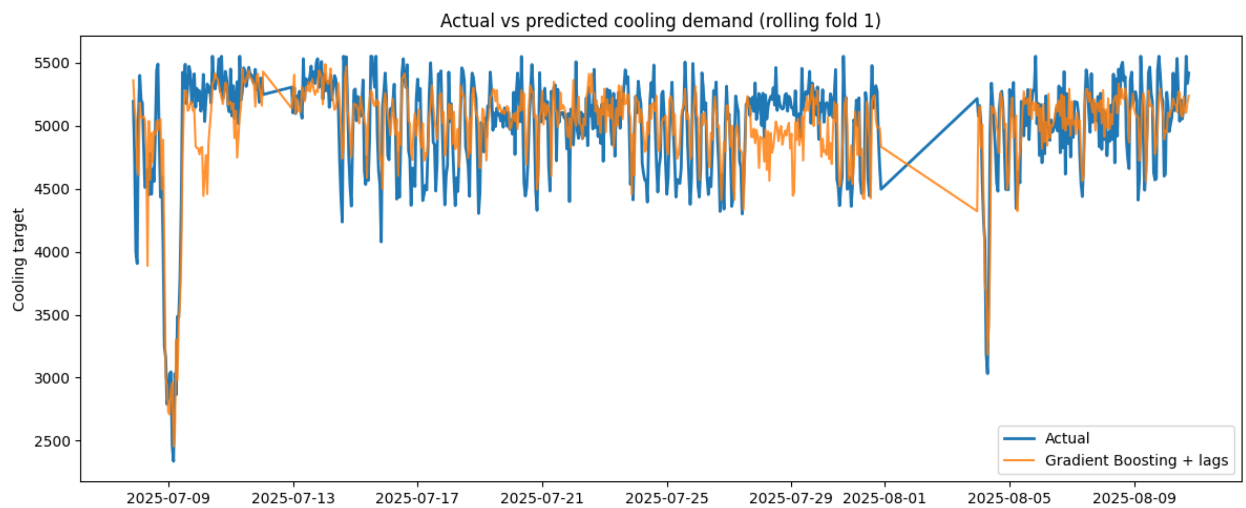

Fig. 13. Comparison between the cooling-demand target and the expected-demand benchmark

As shown in Fig. 13, the expected-demand benchmark follows the main level and short-term dynamics of the observed cooling-demand target while still preserving visible deviations between actual and expected behaviour. This is precisely the analytical role required from the benchmark in the present thesis. The selected model tracks the operational structure of cooling demand sufficiently well to serve as a credible expected-demand reference, but it does not absorb all residual variation into the forecast itself. The remaining deviations therefore become analytically meaningful rather than being treated only as forecast error.

In methodological terms, the benchmark-based cooling assessment provides the bridge between expected-demand modelling and intelligent control analysis. It identifies where actual cooling departs from expectation and therefore where further control-oriented filtering becomes meaningful. The next section builds directly on this result by converting benchmark deviation into a narrower set of candidate control hours through residual-based eligibility filtering and contextual restriction.

3.3. From benchmark deviation to candidate control hours

The next analytical step was to translate benchmark deviation into a smaller and more operationally meaningful set of candidate control hours. This step is necessary because the benchmark-based cooling assessment in Section 3.2 identifies deviation relative to expected demand, but does not by itself determine whether that deviation is sufficiently stable and contextually plausible to justify intervention. In the present thesis, candidate control hours are therefore not defined by positive residuals alone. They are obtained through a staged filtering process that reduces the hourly timeline from broad model-relative deviation to a narrower set of potentially actionable hours.

The starting point of this process was the bias-corrected residual between actual and expected cooling demand. Positive residual values indicate hours in which observed cooling remained above the expected-demand benchmark after correction for systematic model bias. However, the thesis does not interpret every such hour as a meaningful control opportunity. A positive residual may still reflect one-off fluctuation, local forecasting mismatch, or an operational condition in which intervention would not be analytically defensible. For that reason, the residual signal was used only as the first screening layer.

The first reduction step was seasonal restriction. Candidate control hours were limited to the warm season, because this is the period in which cooling demand is structurally most relevant and where benchmark-relative excess can be interpreted most plausibly in control-oriented terms. Outside this seasonal window, positive residuals were retained for descriptive interpretation but not treated as the basis for later intervention logic. This restriction therefore narrowed the deviation analysis from a full-year benchmark comparison to the part of the annual cycle most relevant for intelligent control assessment.

The second reduction step was persistence filtering. In the present framework, one isolated positive residual hour was not considered sufficient evidence of a meaningful control opportunity. Instead, residual exceedance had to persist over time before an hour could remain eligible for later strategy evaluation. This step reduced the influence of short-lived noise and ensured that the candidate set reflected sustained excess rather than transient fluctuation. The persistence condition is especially important in the present thesis because the methodology is designed to test bounded and context-aware intervention logic rather than reactive pointwise correction of every local forecast deviation.

table 3.

Reduction of the hourly timeline to candidate control hours

Filtering stage	**Hours retained**	**Interpretation**
Warm-season hours	3537	Seasonally eligible analysis window
Positive residual hours	1832	Hours where actual cooling exceeded the expected-demand benchmark after bias correction
Meaningful excess hours	458	Hours exceeding the selected residual threshold
Eligible persistent hours	38	Final candidate set retained for later policy evaluation

As shown in Table 3, the candidate control set represents a strongly filtered subset of the warm-season timeline, which confirms that the later intelligent control analysis is based on selective admissibility rather than on broad residual thresholding alone.

Taken together, these filters transformed the benchmark deviation signal into a reduced candidate-hour set for later policy evaluation. This filtering stage is analytically important because it marks the transition from descriptive benchmark comparison to control-oriented admissibility. In other words, Section 3.2 established where actual cooling differed from expected cooling, while the present section establishes which of those differences are sufficiently selective and sustained to function as candidate intervention opportunities.

The resulting reduction in eligible hours also clarifies why the thesis does not interpret benchmark-relative excess at annual scale as evidence of large and continuous overcooling. The aggregate benchmark balance was close to neutral, but this does not preclude the existence of narrower windows of locally concentrated and potentially actionable excess. The candidate-hour logic is therefore intentionally selective. It is designed to identify smaller, more defensible control opportunities rather than to maximize the apparent amount of reducible cooling.

This reduction stage also provides the link between benchmark assessment and the later strategy hierarchy. At this point in the analysis, the hourly timeline has been narrowed to a candidate set defined by benchmark-relative excess, warm-season admissibility, and persistence. The following section then evaluates how different policies within the B0-S3 family act on this candidate space and how much additional selectivity is introduced by load awareness and operating-mode or risk-state context.

3.4. Operating modes, risk states, and temporal control behavior

After the candidate control hours had been identified through benchmark-based deviation and eligibility filtering, the next analytical step was to examine the contextual structure in which those hours occurred. In the present study, this contextual layer was represented through operating modes and their interpretation as intervention-risk states. The purpose of this stage was not to repeat the clustering methodology described in Chapter 2, but to show what the retained modes meant in practical terms and why they mattered for later strategy comparison.

The operating-mode layer was introduced because benchmark-relative excess does not have the same practical meaning under all operating conditions. A positive deviation observed under lighter load and milder thermal context may be more compatible with bounded intervention than a similar deviation observed under heavier load or more demanding environmental conditions. The operating modes therefore function as a contextual interpretation layer that refines the admissibility of later control actions.

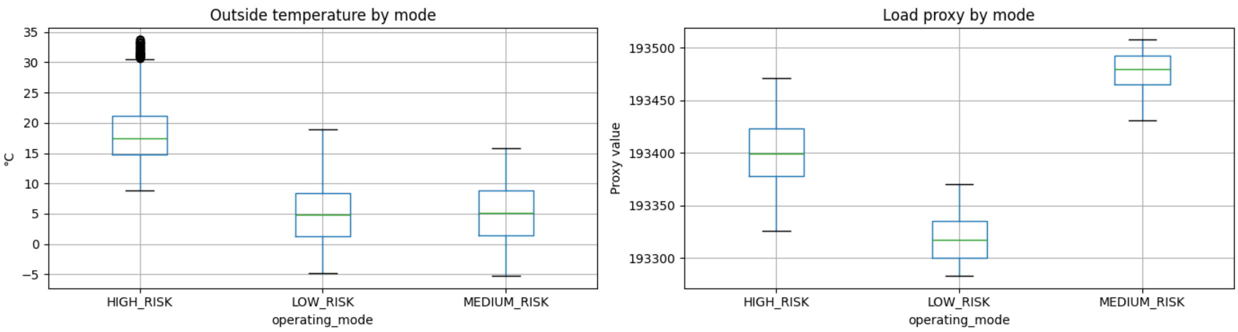

Fig. 14. Operating-mode differences in outside temperature and load proxy

As shown in Fig. 14, the operating modes correspond to distinguishable environmental and operational contexts. The mode distributions differ systematically in terms of outside temperature and load-proxy level, which supports the interpretation that the retained groups reflect meaningfully different conditions of site operation. This is important because it shows that the later risk-state logic is grounded in observable context rather than imposed independently of the data.

The environmental dimension of this interpretation is visible in the distribution of outside temperature across modes. The retained groups occupy different thermal ranges rather than fully overlapping around one common median. At the same time, the load-proxy distributions show that the modes are also separated by operational intensity. In other words, the retained states distinguish not only warmer from cooler hours, but also lighter from heavier site-operation conditions.

Taken together, these two dimensions justify the later interpretation of the modes as low-, medium-, and high-risk intervention states. The low-risk state corresponds to operating conditions in which the joint load-thermal context is most compatible with bounded cooling reduction. The medium-risk state represents an intermediate context in which intervention may still be plausible, but with greater caution. The high-risk state corresponds to the most constrained operating context, where the combination of environmental and operational conditions makes bounded intervention least defensible.

This interpretation becomes especially important in the later strategy comparison. The simpler strategies operate mainly through benchmark deviation and persistence logic, whereas the richer strategies narrow the intervention space further by incorporating contextual admissibility. The operating-mode layer therefore provides the main bridge between residual-based candidate-hour identification and more selective control behaviour.

Overall, the results in Fig. 14 show that the retained operating modes correspond to structured and interpretable differences in outside temperature and operational load. This supports their use as intervention-risk states in the control framework and provides the necessary contextual basis for the comparison of the B0-S3 strategy family in the next section.

3.5. Strategy family B0-S3 and comparative control results

After the contextual structure of operating modes and intervention-risk states had been established, the next analytical step was to compare how the different control policies behaved when applied to the candidate control-hour set. This comparison was carried out through the strategy family B0-

S3 introduced in Chapter 2. The purpose of the comparison was to examine how increasing contextual intelligence changes selectivity, action frequency, and the practical defensibility of intervention.

The four strategies form a progression from simple residual-based action logic to richer context-aware control. B0 functions as the lowest-complexity baseline. It applies the residual-threshold rule to the eligible state without additional contextual refinement and therefore represents the broadest and least selective policy in the family. S1 adds persistence and acts as the forecast-based benchmark strategy. S2 extends this logic with load awareness and allowed-hour logic, while S3 further incorporates the operating-mode or risk-state layer. In this way, the strategy family allows the thesis to compare not only whether benchmark-based intervention opportunities exist, but also whether those opportunities remain defensible when progressively stricter contextual admissibility is imposed.

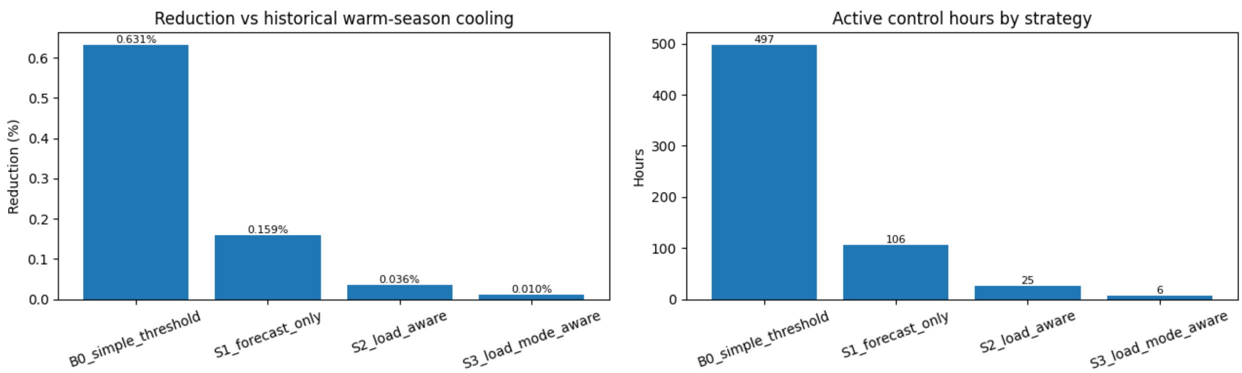

Fig. 15. Comparative control results for the B0-S3 strategy family

Fig. 15 shows a clear trade-off between reduction magnitude and selectivity. B0 produced the largest aggregate effect, with a reduction of 0.631270% relative to historical warm-season cooling and 497 active control hours. S1 reduced warm-season cooling by 0.158979% with 106 active control hours. S2 reduced warm-season cooling by 0.035591% with 25 active control hours. S3, the richest and most selective policy, reduced warm-season cooling by 0.010249% with only 6 active control hours. This progressive decline in both reduction and action count should not be interpreted as methodological weakness. It shows that once richer contextual admissibility conditions are imposed, the margin for bounded intervention becomes much narrower and more selective.

The strategy comparison shows that the main analytical effect of adding load awareness and operating-mode context is not to maximize gross reduction, but to concentrate intervention in a smaller and more defensible subset of the warm-season timeline. In this sense, the richer strategies do not fail because they reduce less. They reveal that a large share of the apparent intervention space identified by simple residual logic becomes analytically questionable once broader operating context is taken into account.

Aggregate comparisons show how often each strategy acts and how much reduction it produces, but they do not fully show how the control logic behaves within an individual operating episode. For this reason, Fig. 16 presents a representative 24-hour example of a simulated S3 intervention window. The figure compares historical actual cooling, expected cooling demand, and simulated controlled cooling, while the shaded interval marks the hours identified by the control logic as eligible for intervention.

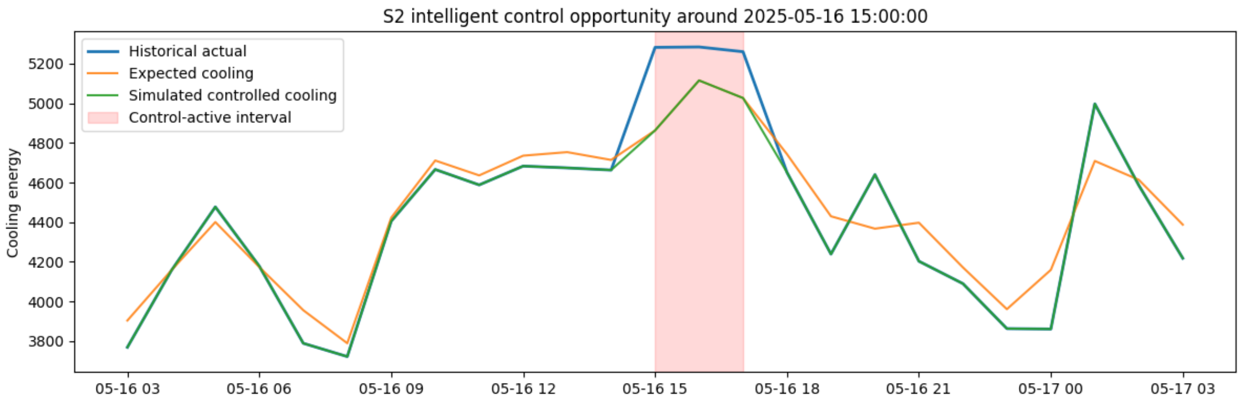

Fig. 16. Example of a 24-hour simulated S3 control opportunity

The example is important because the proposed intelligent control framework operates on hourly data, but its decisions are not interpreted as isolated one-hour events. Eligibility depends on the sequential interaction of several conditions, including positive residual behaviour, warm-season admissibility, persistence, and contextual risk-state constraints. A purely aggregate presentation of active hours and total reduction does not fully show how these conditions combine in time. The 24-hour view therefore provides an interpretable operational-scale illustration of the control logic.

In the example shown, the expected cooling benchmark remains below the historical actual series during the control-active interval, indicating a period of positive excess relative to expected demand. Once the intervention conditions are satisfied, the simulated controlled cooling series is shifted downward in a bounded way rather than forced toward the benchmark or below it. This reflects the design principle of the framework: the method does not attempt actuator-level optimisation, but instead identifies context-aware opportunities for cautious supervisory reduction.

The use of a 24-hour window is deliberate. A single-hour illustration would be too narrow to show persistence and pre-activation context, while a much longer horizon would make the intervention logic visually difficult to interpret. The day-scale window therefore provides a practical compromise. It is long enough to show the development of the excess-cooling episode and the bounded response, but still compact enough to remain analytically readable.

The comparison is especially important when interpreted alongside the contextual results of the previous section. B0 and, to a lesser extent, S1 act on a relatively broad candidate space defined mainly by deviation and persistence. S2 narrows this space by requiring compatibility with the selected load proxy, and S3 becomes the most selective policy because it also requires compatibility with the operating-mode or risk-state layer. The later strategies therefore do not simply reduce action frequency arbitrarily; they apply increasingly strict admissibility logic to the same benchmark-based candidate framework.

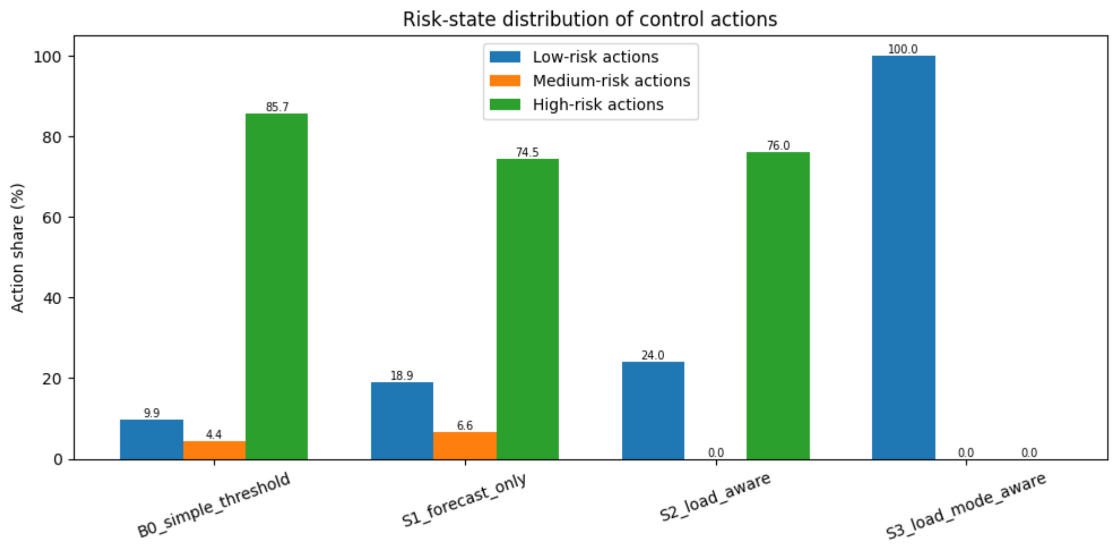

Fig. 17. Distribution of control actions in intervention-risk states for the B0-S3 strategy family

Fig. 17 shows that the richer strategies increasingly concentrate their actions in more intervention-compatible contexts. This is particularly important for S3, because its small number of retained actions reflects strong contextual selectivity rather than random reduction in policy activity. In other words, the strategy family does not only differ in how much it acts, but also in where it acts within the operating-risk structure of the site.

The most important result is not that one strategy maximises reduction in absolute terms, but that the strategy family reveals a structured trade-off between breadth and defensibility. B0 identifies the widest intervention margin, but it does so with limited contextual discrimination. S1 introduces a more realistic forecast-based logic through persistence. S2 and S3 progressively narrow the intervention space, but the remaining actions become more strongly aligned with contextual admissibility. This supports the main argument of the thesis that intelligent cooling control should be understood not only as maximising apparent savings, but as identifying bounded and context-aware opportunities for improvement under real telemetry constraints.

The comparative strategy results show that benchmark-based intelligent control is feasible in analytical terms, but that the amount of intervention judged plausible decreases sharply once richer contextual logic is imposed. This finding is not a limitation of the methodology; it is one of its key results. It shows that context-aware control is more selective and therefore potentially more defensible than broad threshold-based intervention. design choices of the control framework are varied in sensitivity analysis.

3.6. Sensitivity and robustness analysis

The final analytical step was to test whether the main control conclusions remained stable when the design choices of the framework were varied. This stage was necessary because the proposed intelligent control methodology is not a closed-form optimized controller, but a structured analytical framework built from several modelling and policy choices. For that reason, the results should not be interpreted only under one exact baseline specification. Instead, they should be examined under plausible alternative settings in order to determine whether the observed strategy hierarchy and the broader control conclusions are robust.

In the present study, sensitivity analysis is interpreted comparatively rather than exhaustively. The objective is not to search for one globally optimal control specification, but to test whether the main analytical findings persist when the framework is made somewhat stricter, somewhat more permissive, or contextually redefined. A robust result in this setting is one in which the general conclusions remain stable: the simpler strategies act more broadly, the richer strategies act more selectively, and the site does not exhibit evidence of large broadly reducible cooling demand under conservative context-aware logic. This broader stability is already visible in the experiment summary, which shows that proxy sensitivity, cluster-number sensitivity, and supervisory-parameter sensitivity all support the same overall conclusion. Extended sensitivity-analysis outputs for alternative thresholds, persistence rules, reduction caps, and cluster numbers are reported in Annex 8.

The first sensitivity dimension concerned the choice of load proxy used in the context-aware strategies. This was an especially important test because the load proxy enters both the S2 policy and the operating-mode layer used in S3. If the main control results changed radically under alternative proxy definitions, then the interpretation of context-aware intervention would become much weaker. The thesis examined whether the observed control hierarchy remained stable when different operational-intensity proxies were considered.

As described in Chapter 2, candidate operational proxies were compared on four criteria: completeness, variability, relation to actual cooling demand, and practical interpretability as indicators of site operating intensity. The candidate signals retained for direct comparison in the control-extended dataset were 80, 188, and 179. All three had complete overlap with the control dataset, but they behaved differently in terms of correlation structure and contextual usefulness. Signals 80 and 188 showed the strongest raw correlation with actual cooling demand, with correlations of 0.852726 and 0.851444 respectively, whereas signal 179 showed a much weaker direct correlation of 0.043440.

The sensitivity question was not simply which signal tracked actual cooling most closely. A proxy too close to direct cooling behaviour risks weakening the interpretability of the contextual layer. For this reason, the final baseline proxy was not selected by correlation alone. Signal 179 was retained because it provided a more useful operational-intensity variable for control-state design and clearer separation in the operating-mode layer, whereas the more strongly correlated alternatives behaved too much like direct cooling-state indicators.

The proxy-sensitivity results show that the overall control hierarchy remained stable under alternative load-proxy formulations. Simpler threshold-based strategies still identified the broadest intervention set, while richer strategies remained more selective and produced smaller bounded reductions. Thus, changing the operational-intensity proxy influenced the exact shape of the contextual layer, but did not alter the central conclusion: only a small number of lower-risk intervention opportunities remained once conservative context-aware logic was imposed.

This is a useful robustness result because it shows that the thesis does not depend on one specific load-proxy definition. The next subsection therefore examines the second contextual robustness dimension: whether the results remain stable when the operating-mode layer is defined with alternative cluster counts.

4. DISCUSSION

This chapter interprets the main findings of the thesis in relation to the research questions, the methodological position of the study, and the practical problem of cooling-energy analysis in a telecom central office. The discussion focuses on what the findings mean analytically and why they matter. Particular attention is given to the role of target engineering under fragmented telemetry, the function of the expected-demand benchmark as an enabling layer, and the implications of the intelligent control results for context-aware cooling analytics. The chapter then identifies the methodological contribution of the thesis, outlines its practical implications, and discusses the main limitations and directions for future research. The code developed for data preparation, target engineering, forecasting experiments, and intelligent control evaluation is available in the public GitHub repository mentioned in Annex 9.

4.1. Main findings in relation to the research questions

The findings of the thesis should be interpreted first in relation to the main research question, which asked to what extent intelligent cooling control strategies can identify opportunities to improve cooling energy use in a telecom central office environment. The results show that such opportunities can be identified, but that they are relatively narrow and strongly dependent on contextual admissibility. At full-year scale, the benchmark-based comparison did not indicate large systematic overcooling. However, the warm-season analysis revealed smaller windows of positive deviation that could be interpreted as candidate control opportunities once residual, seasonal, and persistence conditions were applied. The comparison of the B0-S3 strategy family then showed that these opportunities shrink sharply as the control logic becomes more context-aware. This is the most important discussion point of the thesis. It means that intelligent cooling control in this case is best understood not as a search for large broad reductions, but as a method for isolating a small number of bounded and operationally defensible intervention opportunities. The richer strategies are therefore not weaker because they reduce less. Their lower reduction is itself a result, because it shows how much of the apparent intervention space disappears once load context and operating-mode or risk-state admissibility are taken seriously.

The second research question asked which forecasting approach provides the most reliable estimate of expected cooling demand for intelligent cooling control. The results show that one-hour-ahead cooling demand at the selected site was strongly persistence-driven, while still benefiting from nonlinear lag-aware modelling and contextual variables. The strongest benchmark candidates were the lagged tree-based models, with Gradient Boosting retained as the final expected-demand benchmark and Random Forest remaining the closest alternative. This finding is important, but it should be interpreted in its proper role within the thesis. Forecasting was not the final objective of the study. Its importance lies in the fact that the selected benchmark enabled expected-versus-actual cooling assessment and therefore made benchmark-based intelligent control possible. In other words, the contribution of the forecasting stage is not only that it identified a high-performing model, but that it established a credible expected-demand reference from which later control-relevant deviation could be defined. This supports the broader methodological position of the thesis: expected-demand modelling is a necessary enabling layer for intelligent control, but not the main contribution in itself.

The third research question asked which operational and environmental variables are most relevant for constructing a control-oriented analytical basis for intelligent cooling control. The results show that such a basis can indeed be constructed from fragmented telecom telemetry, but only through explicit methodological structuring. The most important elements were the engineered hourly cooling-demand target, lagged demand history, selected internal operational variables, external weather variables, seasonal admissibility, persistence logic, a load proxy, and a thermal-context representation. This finding is significant because it shows that the analytical basis for intelligent control does not emerge automatically from the monitoring environment. Instead, it must be built step by step through signal screening, target engineering, dataset design, and contextual variable selection. In this sense, the thesis supports the view that in telecom cooling analytics, the transformation of fragmented monitoring data into a usable control state is itself a substantial research task rather than a minor technical preparation step.

Taken together, these findings answer the research questions in a coherent order. First, the study shows that intelligent cooling control opportunities can be identified, but only in a narrow and context-dependent way. Second, it shows that this identification depends on the availability of a reliable expected-demand benchmark, which in the present case was best provided by a lagged nonlinear tree-based model. Third, it shows that the entire control logic depends on an explicitly engineered analytical basis constructed from fragmented operational and environmental data. The overall implication is that intelligent cooling control in a telecom central office is feasible under real telemetry constraints, but it must be understood as a benchmark-based, context-aware, and bounded analytical framework rather than as a claim of large directly deployable savings.

4.2. Methodological contribution of the thesis

The methodological contribution of the thesis lies in the integration of analytical layers that are often treated separately in the literature. Existing telecom cooling studies are comparatively strong in monitoring, KPI design, and descriptive diagnostics, while forecasting studies typically focus on predictive accuracy and advanced-control studies usually assume richer state feedback, validated control variables, or deployable optimisation settings. By contrast, the present thesis develops a single control-oriented framework for analysing cooling-energy behaviour under real telecom data constraints. The contribution is therefore not a new forecasting algorithm or a new control law in isolation, but an integrated method for evaluating cooling-control opportunities under incomplete telemetry.

A second contribution is the explicit treatment of target engineering as a core research task. In the present case, cooling demand was not directly available as a ready-made hourly variable and had to be reconstructed from fragmented cumulative subsystem measurements. The thesis therefore shows that in real telecom monitoring environments, the validity of later modelling and control interpretation depends strongly on this earlier engineering step. In this sense, the study contributes a reproducible approach to moving from nominal signal availability to a defensible cooling-demand target and control-state representation.

A third contribution lies in the control methodology itself. The thesis formulates intelligent cooling control as a simplified sequential decision framework built around state, policy, action, objective, and historical outcome. This framing is methodologically important because it positions the study between pure forecasting and full advanced control. Rather than overclaiming full model predictive control, reinforcement learning, or formal Markov decision process optimisation, the framework shows how benchmark-relative deviation, persistence, load context, and operating-mode or risk-state information can be combined into a bounded and context-aware control logic under incomplete telemetry.

Finally, the study contributes a comparative logic for intelligent control interpretation. Rather than presenting one single control rule as the result, it develops and tests the B0-S3 strategy family as a hierarchy of increasingly context-aware policies. This makes it possible to show not only whether control opportunities exist, but how their interpretation changes when richer admissibility conditions are introduced. The main methodological value of this design is that it reveals the trade-off between breadth and defensibility in a transparent way.

4.3. Practical implications for telecom cooling analytics

The practical implications of the thesis lie primarily in decision support rather than in immediate direct deployment. The proposed framework shows that benchmark-based cooling analytics can be used to move beyond descriptive monitoring toward structured identification of possible intervention windows. In a telecom central office, this means that operational data can support not only retrospective reporting of cooling demand, but also the detection of hours in which cooling behaviour appears elevated relative to what would be expected under the observed conditions. This is a useful practical shift because it turns monitoring data into an analytical basis for more targeted operational review.

A second implication is that cooling-control support should be introduced gradually rather than as an all-or-nothing transition to advanced control. The results of the strategy family show that simple residual-based logic identifies a broad intervention margin, but that this margin becomes much smaller when persistence, load context, and operating-mode admissibility are introduced. In practical terms, this suggests that telecom cooling analytics could first be implemented as a layered supervisory-support tool: beginning with benchmark-relative alerts, then adding persistence logic, and only later incorporating richer contextual screening. This staged implementation path is more realistic than assuming that a site can move directly from fragmented monitoring telemetry to fully optimised closed-loop control.

The thesis also suggests that expectations about cooling reduction should remain cautious. The results do not support the idea of large broadly available savings once context-aware admissibility is imposed. Instead, they indicate that the most defensible intervention opportunities are relatively narrow and selective. Practically, this means that the value of intelligent cooling analytics in telecom environments may lie less in identifying large aggregate savings and more in helping operators detect a small number of recurring, benchmark-relative inefficiencies that would otherwise remain hidden within normal load variation.

Taken together, the practical value of the thesis lies in providing a benchmark-based and context-aware analytical framework that can support more selective operational review of cooling behaviour in telecom environments.

4.4. Limitations and future research

Several limitations of the thesis should be acknowledged when interpreting the results. The first concerns data availability and state observability. Although the study demonstrates that fragmented telecom telemetry can be transformed into a control-oriented analytical basis, the source environment did not provide direct indoor room-temperature measurements, richer internal thermal-state feedback, or actuator-level command variables. As a result, the thesis cannot claim full physical observability of the cooling process and does not support direct closed-loop optimisation in the sense of full model predictive control. The benchmark-based and context-aware control logic developed here should therefore be interpreted as an analytical and historically evaluated framework rather than as a deployable controller.

A second limitation concerns the target itself. The hourly cooling-demand series used throughout the study was engineered from cumulative subsystem measurements rather than observed as a native ready-made cooling-load variable. This was a necessary and methodologically transparent choice under the available telemetry conditions, but it also means that all later forecasting and control results depend on the validity of that reconstruction. Although the target engineering process was designed carefully and reviewed for plausibility, the resulting series remains an analytical approximation of site-level cooling demand rather than a direct ground-truth measurement.

A third limitation concerns the case-study scope. The thesis is based on one selected telecom central office and one annual study window. This design was appropriate for methodological depth and traceability, but it limits broader empirical generalisation. The results should therefore not be interpreted as universally representative of all telecom central offices or all cooling environments. In particular, the balance between persistence-driven cooling behaviour, weather sensitivity, and intervention admissibility may differ in sites with other technical configurations, metering practices, or climatic conditions.

A fourth limitation concerns the benchmark-based interpretation itself. The selected expected-demand model provides a strong analytical reference, but it remains a model rather than a physical truth. Positive residuals therefore indicate potential excess cooling relative to the selected benchmark, not direct proof of physical overcooling. The control results must therefore be interpreted as benchmark-relative and context-aware opportunities rather than guaranteed physically safe reduction actions.

A final limitation concerns the historical nature of the control evaluation. The B0-S3 strategies were assessed retrospectively on historical data rather than tested in live operation. This means that the results show what the control logic would have identified and how much bounded reduction it would have implied under the chosen assumptions, but not how the real site would have responded dynamically to those interventions.

These limitations also define the main directions for future research. The first priority is richer site telemetry, especially direct indoor thermal-state measurements and actuator-level control variables. The second is broader empirical testing across multiple telecom sites to assess transferability under different infrastructure and climate conditions. The third is online or semi-online validation, where the benchmark-based control logic could be tested in operational monitoring systems as a decision-support tool. The fourth is movement toward more formal predictive-control implementations once richer state feedback and validation conditions become available.

Taken together, these limitations do not invalidate the results of the thesis. Rather, they define the boundaries within which the findings should be interpreted and clarify the next steps needed to move from benchmark-based intelligent cooling analytics toward more advanced control support.

CONCLUSIONS

The aim of this thesis was to develop and evaluate intelligent cooling control strategies for a telecom central office under real telemetry constraints. More specifically, the study examined to what extent intelligent cooling control strategies can identify opportunities to improve cooling energy use when the analytical basis must first be constructed from fragmented monitoring data rather than taken as ready-made. This aim was pursued through a sequential analytical framework combining source qualification, signal screening, cooling-demand target engineering, expected-demand benchmark selection, benchmark-based cooling assessment, and context-aware intelligent control evaluation. Based on the results of the study, it can be concluded that the aim of the thesis was achieved within the defined scope.

The tasks set in the thesis were also completed within the defined scope. The study successfully identified and qualified a suitable site and signal basis, reconstructed an hourly cooling-demand target from fragmented subsystem telemetry, developed and compared expected-demand forecasting models, selected a benchmark model for expected-versus-actual assessment, and evaluated bounded intelligent control strategies under historical operating conditions. In this sense, both the overall aim and the individual methodological tasks of the thesis were achieved.

The first and main research question asked to what extent intelligent cooling control strategies can identify opportunities to improve cooling energy use in telecom central office environments. The results show that such opportunities can be identified, but that they are limited in scale and become substantially narrower when stricter contextual admissibility is imposed. The annual benchmark-based assessment did not support large-scale systematic benchmark-relative overcooling at the selected site. At the same time, the expected-versus-actual comparison revealed smaller warm-season windows of positive deviation that could be interpreted as candidate control opportunities. These windows were then filtered through persistence, load context, and operating-mode or risk-state logic. The comparative strategy analysis showed that the broadest strategy, B0, identified the largest apparent intervention margin, whereas the richer strategies S1, S2, and especially S3 reduced this margin sharply by enforcing progressively stricter conditions. The thesis therefore concludes that intelligent cooling control in this setting should be interpreted as bounded and context-aware decision support rather than as a claim of large directly available cooling reductions.

The second research question asked which forecasting approach provides the most reliable estimate of expected cooling demand for intelligent cooling control. The comparative modelling experiment showed that short-horizon cooling demand at the selected site is strongly persistence-driven, but that persistence alone is not sufficient as the final expected-demand benchmark. Among the evaluated alternatives, lagged nonlinear tree-based models performed best, with Gradient Boosting selected as the final expected-demand benchmark and Random Forest retained as the strongest alternative candidate. Classical autoregressive models and simpler lagged baselines remained useful as methodological references, but they did not provide the same benchmark quality under the chosen validation framework. The thesis therefore concludes that the most reliable expected-demand estimate in the present case was obtained from a lagged nonlinear benchmark model rather than from a simple persistence baseline or a purely classical time-series approach. Forecasting was thus important not as a separate endpoint, but as the enabling layer that made benchmark-based cooling assessment and later control logic possible.

The third research question asked which operational and environmental variables are most relevant for constructing a control-oriented analytical basis for intelligent cooling control. The thesis showed that such a basis can be built from fragmented telecom telemetry, but only through explicit methodological structuring. The most important analytical components were the engineered hourly cooling-demand target, lagged target history, selected internal operational telemetry, external weather variables, seasonal eligibility, persistence logic, a load proxy, and a thermal-context representation. The study therefore concludes that the transformation of fragmented operational and environmental data into a control-oriented analytical basis is feasible, but it is not automatic. It requires careful signal screening, cross-table mapping, measurement-layer qualification, target engineering, and contextual variable design. In this sense, the thesis shows that the analytical basis for intelligent control in telecom cooling is itself a substantial research task.

Taken together, the answers to the three research questions form one coherent overall result. The study showed that fragmented telemetry can be transformed into a usable control-oriented analytical basis, that a reliable expected-demand benchmark can be selected and validated from that basis, and that this benchmark can then be used to identify a small but meaningful set of context-aware intervention opportunities through a bounded intelligent control framework.

The main contribution of the thesis lies in the development of an integrated methodological framework for telecom cooling analytics under incomplete telemetry conditions. This framework combines target engineering, expected-demand benchmark modelling, benchmark-based cooling assessment, and intelligent control evaluation within one applied case study. Rather than claiming deployable model predictive control or reinforcement-learning control, the thesis formulates intelligent control as a simplified sequential decision framework based on state, policy, action, objective, and historical outcome. It also contributes a comparative way of interpreting intelligent control through the B0-S3 strategy family, showing that the strength of an intelligent control methodology is not necessarily reflected in the largest estimated reduction, but in the credibility and selectivity of the actions it retains.

In practical terms, the thesis suggests that telecom cooling analytics should move beyond descriptive monitoring toward benchmark-based and context-aware decision support. At the same time, it shows that expectations about reduction potential should remain cautious. The results do not support large immediate claims of savings under conservative conditions. Instead, they support the use of intelligent control analytics as a tool for identifying narrower and more defensible intervention windows that can later be reviewed, validated, and potentially tested in more operational settings. Overall, the thesis shows that intelligent cooling control in a telecom central office is feasible as a benchmark-based and context-aware analytical framework under real telemetry constraints, and that its value lies in providing a structured way to identify, interpret, and prioritise smaller intervention opportunities under operational context.

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ANNEX 1. INTELLIGENT CONTROL STRATEGY FAMILIES

table 1A.

Intelligent control strategies reviewed for cooling systems and relevance to the thesis

Strategy family	Main idea and requirements	Relevance to the thesis
Rule-based control	Predefined thresholds, schedules, or logic; requires only basic measurements and domain rules	Useful practical benchmark and conceptual baseline; common in real operation
Predictive control (MPC)	Uses a model, forecasts, constraints, and manipulated variables to optimize future control actions	Core intelligent-control literature anchor, but full implementation requires richer states and actuator-level data than available here
Data-driven / approximate predictive control	Replaces or simplifies physical models through machine-learning models or learned MPC-like policies; lowers modelling burden but still needs structured control setup	Most relevant methodological bridge from forecasting to control; supports supervisory logic derived from predictive models
Model-free predictive control / reinforcement learning	Learns control behaviour directly from data or interaction without an explicit plant model	Reviewed for completeness, but less suitable here because it typically requires richer feedback, training environment, and safety assurance
Cooling-specific predictive control in technical infrastructure	Applies predictive control directly to cooling supply, airflow, or thermal balancing under infrastructure-specific constraints	Closest transfer literature to telecom cooling; shows what full cooling-specific control looks like when zone states and control variables are available

ANNEX 2. SOURCE LAYERS AND ANALYTICAL ROLES

table 2A.

Source layers and their analytical roles in the study

Source layer	**Main tables**	**Main role in the study**
Site and support layer	monitoring_site, monitoring_site_model, swenex_file_data	Site identification, site-model context, supporting utility information, and initial candidate-site screening
Signal metadata layer	enmos_datapoint, datapoint, measuring_point	Signal identification, metadata interpretation, cross-table mapping, and definition of the candidate signal universe
Measurement layer	measurement, measurement_all	Time-stamped signal values used for signal profiling, hourly aggregation, target construction, and dataset assembly
Visible but not used in the final workflow	mobnot, mobnot_reported_state	Present in the accessible reporting layer, but not used in the final analytical pipeline

ANNEX 3. SHORTLISTED COOLING TARGET CANDIDATES

table 3A.

Shortlisted cumulative cooling target candidates and their system role

Datapoint ID	DP ID	Short description	System role
120	V_K78_RZ	COOL Abwärmenutzung EWZ	Core cooling distribution / heat-recovery link
230	V_K81_RZ	COOL Ringleitung 1	Core cooling distribution
46	V_K82_RZ	COOL Ringleitung 2	Core cooling distribution
49	V_K76_RZ	COOL Kaltwasserperipherie Haupt	Core cooling distribution / periphery core
78	V_K74_RZ	COOL Steigstrang RK	Core cooling distribution
60	V_K71_RZ	COOL KM A Verdampfer	Production-side subsystem
221	V_K84_RZ	COOL Leitung B3 U6	Branch-level distribution
149	V_K83_RZ	COOL Leitung B1 U6	Branch-level distribution
232	V_K77_RZ	COOL Kaltwasserperipherie vor Abwärme	Peripheral / pre-heat-recovery branch
194	V_K75_RZ	COOL FC Umformer	Free-cooling / conversion-related subsystem
1	V_E86_RZ	COOL Ringleitung 1	Infrastructure-adjacent signal
201	V_E08_RZ	E08 Wirkenergie Trafo T9 U200	Infrastructure proxy
159	V_E06_RZ	E06 Wirkenergie Trafo T6 U200	Infrastructure proxy
91	V_E06B_RZ	E06B Blindenergie Trafo T6 U200	Infrastructure proxy

ANNEX 4. FORECASTING MODELS EVALUATED

table 4A.

Forecasting models evaluated in the comparative experiment

Model	Family	Main role in the experiment
Naive	Benchmark	Persistence reference for short-horizon prediction
Seasonal Naive	Benchmark	Seasonal baseline in the initial comparison layer
ARIMA	Classical time-series	Initial autoregressive benchmark
SARIMA	Classical time-series	Initial seasonal autoregressive benchmark
Linear Regression	Statistical benchmark	Simple interpretable baseline in the initial comparison layer
Random Forest	Machine learning	Initial nonlinear benchmark
LSTM	Neural sequence model	Initial deep-learning benchmark
Linear Regression + lags	Statistical lagged model	Interpretable autoregressive benchmark in the lagged setup
Random Forest + lags	Machine learning	Main nonlinear lagged candidate
Gradient Boosting + lags	Machine learning	Main boosting-based lagged candidate
LSTM multivariate	Neural sequence model	Sequence-based multivariate lagged candidate
ARIMAX / SARIMAX + exog	Classical exogenous time-series	Weather-extended classical benchmark

ANNEX 5. CANDIDATE LOAD-PROXY COMPARISON

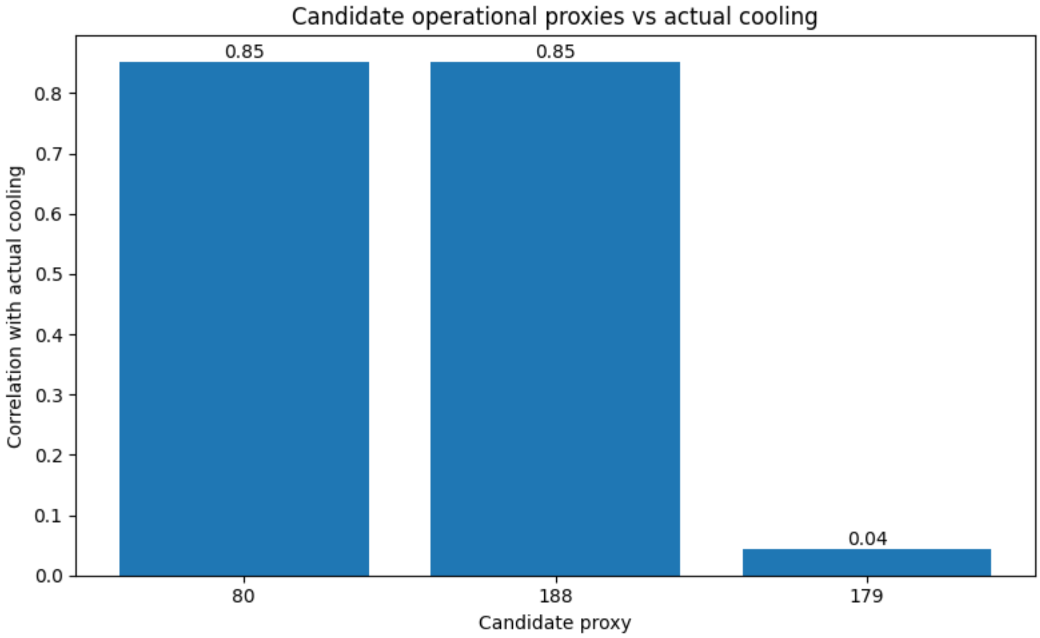

Fig. A3. Candidate operational proxies vs actual cooling

	proxy_col	completeness	mean	std	corr_with_actual	corr_with_residual_bc	warm_minus_nonwarm	day_minus_night
0	80	1.0	9.238145	11.028646	0.852726	0.042977	17.205127	1.432680
1	188	1.0	10.791736	12.920856	0.851444	0.043934	20.110189	1.717449
2	179	1.0	193392.700416	66.073517	0.043440	-0.001708	10.330902	0.144397

Fig. A4. Comparative behaviour of candidate load proxies in relation to cooling-demand context

ts_hour	179	188	80	90
2025-01-01 00:00:00+00:00	193277.15625	2.287222	4.558055	163.666191
2025-01-01 01:00:00+00:00	193277.15625	0.000000	0.000000	169.334431
2025-01-01 02:00:00+00:00	193277.15625	0.000000	0.000000	166.187340
2025-01-01 03:00:00+00:00	193277.15625	0.000000	0.000000	169.021465
2025-01-01 04:00:00+00:00	193277.15625	0.000000	0.000000	164.831135

Fig. A5. Load–thermal context separation for the retained operating-mode structure

ANNEX 6. SENSITIVITY DIMENSIONS

table 5A.

Main sensitivity dimensions in the intelligent control methodology

Component	**Baseline logic**	**Sensitivity alternatives**	**Purpose**
Residual threshold	Meaningful excess defined from the positive residual distribution	Alternative residual quantile cut-offs	Test selectivity of excess-cooling definition
Persistence requirement	Sustained excess required before eligibility	Shorter and stricter persistence rules	Test robustness of temporal admissibility
Reduction cap	Fixed bounded cooling-reduction magnitude	Lower and higher bounded caps	Separate candidate-hour selection from action intensity
Load proxy	Selected operational-intensity proxy	Alternative proxy formulations	Test dependence of context-aware strategies on proxy choice
Operating modes	Three-mode clustering in load-thermal context space	Alternative cluster counts	Test robustness of operating-mode and risk-state definition

ANNEX 7. FULL FEATURE-IMPORTANCE OUTPUTS

	feature	importance_mean	importance_std
0	y_lag_1	390.941680	8.431659
1	90	22.065791	1.989904
2	wx_temp_2m	18.045276	1.338777
3	72	15.625937	1.099950
4	100	12.904101	2.030763
5	35	10.420941	0.713318
6	y_lag_12	7.385311	0.655784
7	14	6.964530	0.932955
8	187	6.485617	0.426581
9	171	6.406678	0.401004
10	108	6.143192	0.802620
11	y_lag_24	4.326473	0.480574
12	wx_dewpoint_2m	2.678281	0.466314
13	202	2.033644	0.258394
14	161	1.749504	0.577678
15	y_lag_168	1.402390	0.285401
16	132	1.364977	0.326276
17	206	1.338365	0.182680
18	178	0.830255	0.255004
19	121	0.809198	0.553581

Fig. A1. Top 20 feature importances for the Gradient Boosting benchmark model

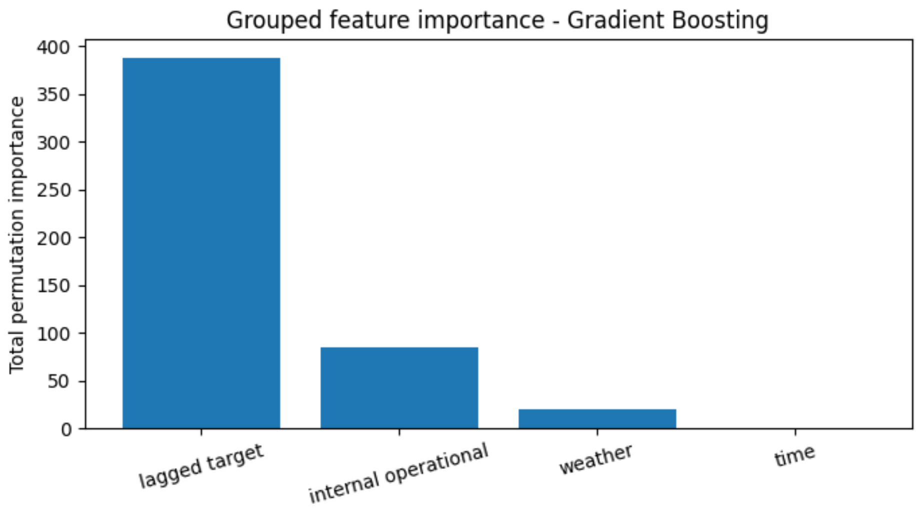

Fig. A2. Grouped feature importance

ANNEX 8. ADDITIONAL SENSITIVITY-ANALYSIS RESULTS

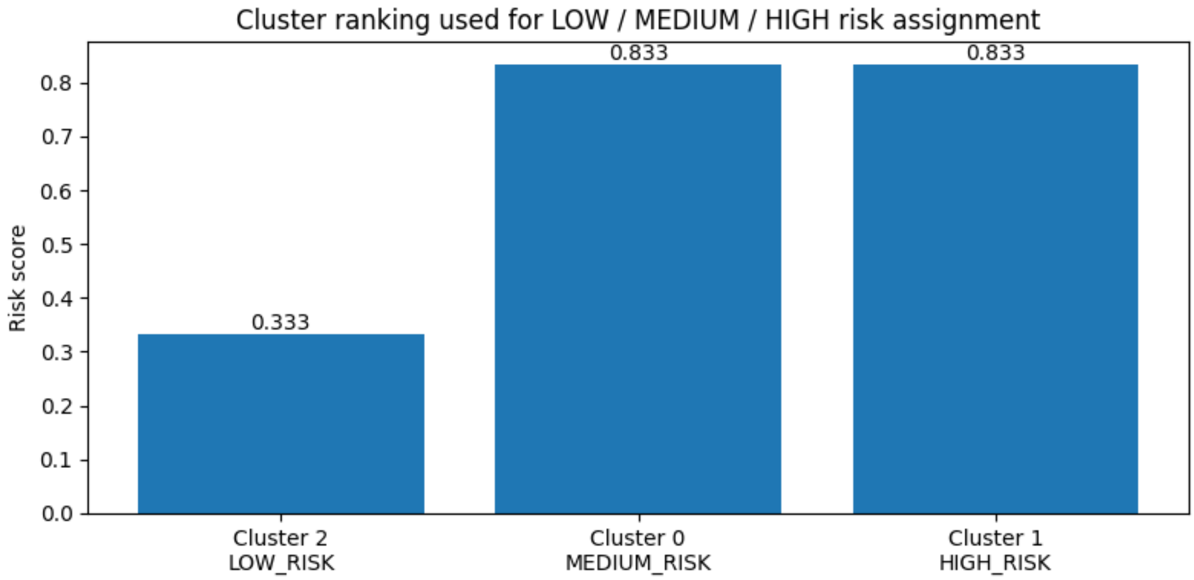

Fig. A6. Cluster ranking used for risk modes

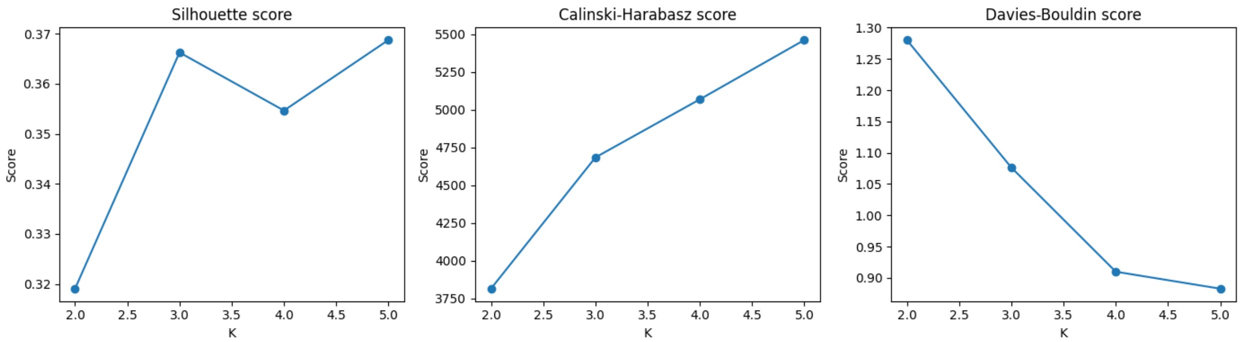

Fig. A7. Cluster count validation

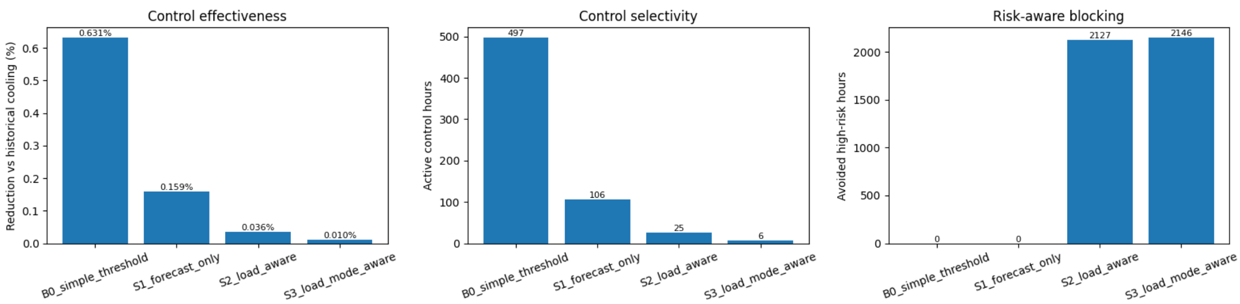

Fig. A8. Control effectiveness, control selictivity and risk-aware blocking

ANNEX 9. CODE AND REPRODUCIBILITY RESOURCES

The analytical code developed for this thesis is documented in a public GitHub repository accompanying the study. The repository contains the notebook-based workflow used for data preparation, target engineering, forecasting experiments, intelligent control evaluation, and generation of supporting analytical outputs. The code repository is provided to improve transparency and reproducibility of the implemented methodology. Repository link is https://github.com/anastasijazu/master-thesis.

Main code file is https://github.com/anastasijazu/master-thesis/blob/main/masters.ipynb. The repository documents the computational workflow used in the thesis, including:

- data preparation and signal-screening logic
- construction of the engineered cooling-demand target
- forecasting-model comparison and benchmark selection
- benchmark-based cooling assessment
- intelligent control strategy evaluation
- generation of selected figures and supporting outputs

The raw monitoring data used in this thesis are not publicly distributed in full because the study is based on real telecom operational telemetry and confidentiality constraints apply. The repository therefore provides the analytical code and workflow structure, but not unrestricted publication of the original source data. As a result, full numerical reproduction depends on access to the underlying data environment or to an appropriately anonymised equivalent dataset.

The GitHub repository should be understood as a reproducibility companion to the written thesis rather than as a substitute for it. The main text remains the primary source for the research design, methodological justification, benchmark-selection logic, intelligent control framework, and interpretation of results.